THE
SCARA Robot

# Dustproof and Dripproof Specification Manual

(Compatible Robot: THE400)
(Compatible Controller: TSL3000/TSL3000E/TS5000-SS)

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Important

- Make sure that this instruction manual is delivered to the final user of Shibaura Machine's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

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# Introduction

Thank you very much for your purchase of Shibaura Machine's robot system.
This manual describes the dustproof and dripproof specifications.
Read this manual before unpacking the product.
Before beginning the work according to this manual, read through the Safety Manual so that you can understand the safety measures.

This manual has the following chapters.

Specifications
Section 1: Specifications
This section describes the basic specifications and names of the units of the dustproof and dripproof robot.

Installation
Section 2: Unpacking and Transportation
This section describes how to unpack the dustproof and dripproof robot and how to transport it to the installation site.
This section also describes precautions for temporarily storing the unpacked robot.

Section 3: Installation of the Robot
This section describes the installation environment, required space, and installation procedure for the dustproof and dripproof robot.

Sections 4 and 5: Setting of the Hand
These sections describe how to install the hand on the dustproof and dripproof robot and how to wire the hand.
Lastly, these sections describe the permissible load conditions of the hand.

Maintenance
Sections 6 and 7: Maintenance
These sections describe how to maintain the units of the dustproof and dripproof robot.

Section 8: Robot Zero Point and Position Detector Error
This section describes the zero point setting procedure for the dustproof and dripproof robot.

Section 9: Replacement Parts
This section describes the replacement parts of the dustproof and dripproof robot.

Section 10: Disposal
This section describes disposal of the robot and parts.

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## Use Range

This product is the horizontally articulated type industrial robot in which the manipulator has two horizontally rotating joints and the mechanical interface on its tip moves vertically and rotates horizontally.
The product has been developed with intention of adapting it to automation of wide range of work including conveying work such as movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product with the aim to adapt it to automation of these work and incorporate it to the automation equipment.

This product is not intended to apply it to work that contacts a human (such as work related to medical care and nursing care), incorporate it to the equipment that supports human life (such as life support device and incubator), incorporate it to the equipment on which a human rides (including transport equipment such as car, aerospace equipment such as airplane, and game equipment such as roller coaster), and incorporate it to the equipment that protects human life and human body (such as safety device).
This product is not intended to incorporate it to the equipment aimed to control large quantity of heat (such as nuclear power controller and incinerator controller).
This product is prohibited from being used to apply it to work or incorporate it to the equipment that is likely to threaten human life and harm a human body directly due to a failure or malfunction. If this product is used, it will be guaranteed under customer's responsibility.

## Use Method

The use conditions for the THE series robot are described. If the product is used in the state that differs from the conditions, unexpected risk may occur. Abide by the conditions for use.

1. Main Specifications of Product: See Specifications "1.4 Specifications Table" and Installation "3.1 Installation Environment."

2. Restrictions on Use: Table 1 to Table 8 give "Restricted Specification Sheet" indicating restrictions on use.

Table 1 Restricted Specification Sheet (1)

| 1. Purpose and application of using product (restrictions on use) | |
|---|---|
| 1) Intended use | Allow the robot to achieve operation along with the purposes (article conveyance, assembly, inspection, machining, etc.).<br>(1) Transportation<br>① Secure the route for transportation to the unpacking site.<br>② Lift the packing box by two or more transportation workers.<br>③ Transport the packing box to the unpacking site.<br>(2) Unpacking<br>① Secure the space required for unpacking.<br>② Unpack the box.<br>③ Make sure that the robot and controller are not damaged.<br>④ Check the accessories and their quantity.<br>⑤ Dispose of or store and manage the packing material.<br>(3) Conveyance<br>① Check the posture of the robot during conveyance.<br>② Check the mounting of the robot clamp.<br>③ Check the tightening of the robot clamp fixing bolts.<br>④ Remove the bolts that secure the robot base.<br>⑤ Lift and convey the robot by two or more workers.<br>(4) Installation<br>① Secure the robot to a frame etc.<br>② Remove the robot clamp.<br>③ Install the controller and perform wiring.<br>④ Install the hand etc. |

Table 2 Restricted Specification Sheet (2)

| 1. Purpose and application of using product (restrictions on use) | |
|---|---|
| 1) Intended use (continued) | (5) Teaching<br>- Arrange a person who observes teaching work outside the dangerous range.<br>- Perform work according to the work regulation.<br>① Use the TP to guide the robot or hold the arm etc. by hand to move the movable part of the robot to any position.<br>② Teach the position where the robot operates.<br>③ Create the operation sequence (program etc.).<br>④ Make sure that signal and data are transferred to and from the external equipment.<br>⑤ Check operation at low speed 8 (250 mm/sec or less).<br>⑥ While increasing the operating speed gradually, make sure that the operation suits the purpose.<br>(6) During operation<br>- A measure has been taken to turn off the servo power when the dangerous range of the robot is approached.<br>- Perform work according to the work regulation.<br>① After making sure that there is no person in the fence, enclosure, etc., start operating the robot.<br>② Make sure that the operation is performed according to the purpose at a predetermined speed.<br>③ Stop operating the robot.<br>(7) Repacking<br>① Remove the hand etc.<br>② Mount the robot clamp.<br>③ Remove the wiring.<br>④ Pack the robot and controller. |

Table 3 Restricted Specification Sheet (3)

| 1. Purpose and application of using product (restrictions on use) | |
|---|---|
| 1) Intended use (continued) | (8) Maintenance<br>- Arrange a person who observes maintenance work outside the dangerous range.<br>- Perform work according to the work regulation.<br>[Work performed with the primary power turned off]<br>① Check the bolts for looseness and backlash and re-tighten them.<br>② Make sure that the cable clamp is not loose.<br>③ Check for damage.<br>④ Clean dust.<br>⑤ Inspect and apply antirust agent.<br>⑥ Check the tension of the timing belt and check for crack on the belt.<br>⑦ Check for wear on the cable and air tube.<br>⑧ Replace the consumable parts for the robot and controller.<br>⑨ Replace the battery for memory in the controller<br>[Work performed with the primary power turned on]<br>① Check the grease state and apply grease (when the servo power is off).<br>① Check the behavior when the robot is moved by jog guiding using the TP (when the servo power is on).<br>② Push each axis by hand and check it for backlash (when the servo power is on).<br>③ Check abnormal vibration and sound during operation (when the servo power is on).<br>④ Check the teaching point for misalignment (when the servo power is on).<br>⑤ Replace the battery for robot position detector (when the servo power is off).<br>(9) Disposal<br>① Mount the clamp on the robot and dispose of the robot using the method defined by each country's laws and regulations. |
| 2) Predictable misuse | - Use of the robot in work which applies excessive impact to the robot or to which excessive impact is applied by the robot<br>- Modification of robot, controller, or optional parts and energization with their cover removed<br>- Operation in other than the specified installation state (floor, suspension, and wall-mounted)<br>- Operation on a frame etc. that do not satisfy the specified strength<br>- Operation under other than the specified installation condition<br>- Entering of the robot into the dangerous range of the robot on the assumption that stop for operation signal wait during operation is operation stop<br>- Use under a condition that exceeds the specification value or use at an improper setting<br>- Stop of the robot due to power-off during operation or while the servo power is on<br>- Unintended operation caused by erroneous pressing of a button<br>- Operation without checking operation in teaching mode<br>- Operation without checking tightening of the robot and hand fixing bolts<br>- Unintended operation caused by erroneous wiring<br>- Unintended operation caused by erroneous program selection<br>- Riding on or leaning against the robot or hanging on the arm |

Table 4 Restricted Specification Sheet (4)

| 1. Purpose and application of using product (restrictions on use) | |
|---|---|
| 2) Predictable misuse | - Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot<br>- Modification of robot, controller, and optional parts and energization with these covers removed<br>- Operation in other than the specified installation state (floor, suspension, and wall-mounted)<br>- Operation on the frame, etc. that do not satisfy the specified strength<br>- Operation under other than the specified installation condition<br>- Entry into the dangerous range of the robot on the assumption that stop for operation signal wait during operation is operation stop<br>- Use under the condition that exceeds the specification value or use in improper setting<br>- Stop of the robot caused by power off during operation and when the servo power is on<br>- Unintended operation caused by erroneous pressing of the button<br>- Operation without checking operation in teaching mode<br>- Operation without checking tightening of the robot and hand set bolts<br>- Unintended operation caused by erroneous wiring<br>- Unintended operation caused by erroneous program selection<br>- Riding on or leaning against the robot or hanging on the arm |
| 3) Unexpected start | - Malfunction caused by excessive noise<br>- Robot start caused by start signal unexpectedly sent from the peripheral equipment to the robot<br>- Unintended start caused by abnormal communication data<br>- Malfunction caused by voltage fluctuation<br>- Power-on caused by energization after power outage<br>- Erroneous start caused by fault |

| 2. Replacement, etc. of program components (restrictions on time) | |
|---|---|
| 1) Mechanical restriction | - Before operating the robot, perform operation at a low speed (20% or less of the maximum speed) for 5 minutes.<br>- Replace consumable parts as required or periodically through daily inspection, regular inspection (every 3 and 6 months and every year) and overhaul (every 5 years). |
| 2) Electrical restriction | |

Table 5 Restricted Specification Sheet (5)

| 3 Movable range of product (restriction on space) | | |
|---|---|---|
| 1) Operating range | Axis 1<br>Axis 2<br>Axis 3<br>Axis 4<br>Additional axis<br>—<br>— | Conforms to 1.4 Specifications Table.<br>Conforms to 1.4 Specifications Table.<br>Conforms to 1.4 Specifications Table.<br>Conforms to 1.4 Specifications Table.<br>Conforms to individual specifications.<br>—<br>— |
| 2) Interface | Transportation: The packed robot shall be able to be conveyed by transportation workers.<br><br>Unpacking: The packed robot shall be able to be carried in.<br>Space of at least 2 $m^3$ shall be secured.<br>Conveyance: The robot shall be able to lifted up and conveyed by two or more workers.<br>There shall be no obstacle on the route from the unpacking site to the installation site.<br>Installation: The robot shall be able to be moved from the unpacking site to the installation site.<br>The product shall be installed at a location suitable for the outer shape and working range of the robot.<br>The product shall be installed at a location suitable for the outer shape and ventilation of the controller.<br>The cable shall be installed at a location that will not be stepped on by workers.<br>Teaching: Teaching work shall be able to be performed from outside the dangerous range of the robot.<br>Space shall be secured to prevent a worker from being caught in the robot and peripheral equipment if an entry is made into the dangerous range.<br>Operation: A fence or enclosure shall be installed to prevent any worker from approaching the dangerous range of the robot.<br>The robot shall be able to be operated from outside the dangerous range of the robot.<br>A worker in the fence, enclosure, etc. shall be visible from the operation site.<br>Repacking: Work shall be able to be performed to remove the hand, mount the clamp, separate the robot and controller, and remove the robot and controller from the frame.<br>The robot shall be able to be moved from the installation site to the packing site. | |

Table 6 Restricted Specification Sheet (6)

| 3 Movable range of product (restriction on space) | |
|---|---|
| 2)<br>Interface | Maintenance: Space required for maintenance shall be secured.<br>Operation after maintenance shall be able to be checked from outside the dangerous range of the robot.<br>In the dangerous range, space shall be secured to prevent a worker from being caught in the robot and peripheral equipment.<br>Disposal: Work shall be able to be performed to remove the hand, mount the clamp, separate the robot and controller, and remove the robot and controller from the frame.<br>The robot shall be able to be moved from the installation site to the disposal location. |
| 3)<br>Work environment | - A fence, enclosure, etc. surrounding the dangerous range of the robot shall be prepared to prevent any person from making an entry during operation.<br>- Display indicating the operation state of the robot shall be prepared to enable a worker to know that the robot is being operated or stopped.<br>- Illumination shall be prepared to enable a worker to be visually confirmed.<br>- The floor shall not be covered with oil etc. to prevent a worker from slipping.<br>- There shall be no noise that impairs attention. |

| 4. Life cycle of product |
|---|
| Among the life cycle of assembly, shipment test, transportation, unpacking, conveyance, installation, teaching, operation, repacking, maintenance and disposal, risk assessment is carried out in the life cycle of “conveyance” (transportation, unpacking, conveyance, installation, and repacking), “use” (teaching, operation and maintenance), and “disposal.” |

Table 7 Restricted Specification Sheet (7)

| 5. Person who approaches the machine | | |
|---|---|---|
| Target | Knowledge and condition | |
| 1) Transportation worker | Knowledge | Contents of the instruction manual |
| | Condition | Person who has received training etc. for work with the conveyance equipment defined by each country's laws and regulations and to whom work has been permitted by the employer |
| 2) Unpacking worker | Knowledge | Contents of the instruction manual |
| | Condition | — |
| 3) Conveyance worker | Knowledge | Contents of the instruction manual |
| | Condition | — |
| 4) Installation worker | Knowledge | Contents of the instruction manual |
| | Condition | — |
| 5) Teaching worker Work observer | Knowledge | Contents of the instruction manual, operation method for target robot, work regulations, and operation of the robot |
| | Condition | Person who has received training etc. for robot teaching work defined by each country's laws and regulations and to whom work has been permitted by the employer |
| 6) Operation worker | Knowledge | Contents of the instruction manual, operation method for target robot, work regulations, and operation of the robot |
| | Condition | — |
| 7) Repacking worker | Knowledge | Contents of the instruction manual |
| | Condition | — |

Table 8 Restricted Specification Sheet (8)

| 5. Person who approaches the machine | | |
|---|---|---|
| Target | Knowledge and condition | |
| 8) Maintenance worker<br>Work observer | Knowledge | Contents of the instruction manual, operation method for target robot, work regulations, and operation of the robot<br>Maintenance method for target robot |
| | Condition | Person who has received training etc. for robot inspection work defined by each country's laws and regulations and to whom work has been permitted by the employer |
| 9) Disposal worker | Knowledge | Contents of the instruction manual and waste treatment method |
| | Condition | — |
| 10) Nearby pedestrian<br>Other workers | - Measures shall be taken to prevent entry into the fence, enclosure, etc. surrounding the dangerous range of the robot.<br>- Measures shall be taken to allow the robot to stop operation when the dangerous range is approached.<br>- If any other worker enters the dangerous range of the robot for work, robot operation shall be stopped to secure safety under observation by the teaching worker or maintenance worker. | |

| 6. Other |
|---|
| |

# Warranty

We conduct a strict tests and inspections before delivery to ensure that the product satisfies our performance standards.
We provide warranty to cover defects of the product based on the following prerequisites.

1. Warranty period The warranty period of the product is either of the following, whichever is earlier.
   1) 18 months after delivery from our factory
   2) 12 months after installation at your site

2. Warranty scope
   1) We provide warranty for the product. Note that only the specifications and functions defined in the product specifications, catalog, and instruction manual are covered by the warranty. Any secondary or incidental damages caused by failure of the product are not covered by the warranty in any circumstances.
   2) We provide repair of a faulty product for free only if the product has been handled and used according to the instruction manual that comes with the product.
      Any repairing after the warranty period passes will be charged.

3. Disclaimer The warranty shall not apply to the following cases.
   1) Any failure or damage arising from your careless use or misuse that is not defined in the instruction manual.
   2) Any failure caused by aging or normal wear and tear (discoloration, deterioration of consumables[*1)].
   3) Any failure regarding sensuous phenomena such as noise not affecting the functions
   4) When any modification or disassembly without our consent is confirmed
   5) Any failure or damage arising from insufficient maintenance and inspection or inappropriate repair
   6) Any failure or damage due to act of God, fires, and other force majeure
   7) Internal data such as programs and points created or modified by users
   8) When a product purchased in Japan is taken abroad

4. Precautions
   1) SHIBAURA MACHINE shall not guarantee the standard performance of the product if you use the product beyond the specifications.
   2) If the customer did not observe the “DANGER,” “WARNING,” and “CAUTION” described in this instruction manual, Shibaura Machine will not assume the responsibility for any consequential accident resulting in injury or death, damage or trouble.
   3) “DANGER,” “WARNING,” “CAUTION” and other descriptions stipulated in this instruction manual are only those which can be assumed by Shibaura Machine without predicting all events in all situations. Please no that the range is limited.

(*1) Consumables refer to the replacement parts for maintenance as listed in “STA1105 THE400 Robot Manual” and “9.1 List of Maintenance Parts.”

## Standards, laws and regulations regarding safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" defines measures for reducing risks regarding machinery.

Step 1: Fundamental safe design measures - Restrictions on force, speed, and energy

Step 2: Safety protection - Installation of protective fences, etc.
Additional protection measures - Installation of emergency stop devices

Step 3: Information for use - Warning labels, alerts, instruction manuals

Based on the above, the ISO/IEC defines the "Guide 51" that groups various standards in a hierarchal structure. Safety standards for industrial robots are standardized as separate safety standards for machinery in layer C.

- ISO 10218, ISO 10218-1

After risk assessment is carried out, provision of residual risk information to the user is standardized.

- IEC 82079-1

Comply with the standards, laws and regulations in the countries where you use industrial robots.

## Safety Precautions

This manual contains important information for the safe use of the robot body and the prevention of injury to the operator and others and the prevention of damage to property.
Read and understand the meanings of the following indications and symbols first, and observe these precautions.

**[Meanings of Indications]**

| Indication | Meaning |
|---|---|
| ⚠ DANGER | Indicates the imminent danger that incorrect operation could cause death or serious injury. |
| ⚠ WARNING | Indicates the danger that incorrect operation could cause death or serious injury. |
| ⚠ CAUTION | Indicates the possibility that incorrect operation could cause injury[1)] to operator. |
| NOTICE | Indicates the possibility that incorrect operation could cause property damage[2)] and advice about performing work. |

1) Injury indicates injuries such as wounds, burns, and electric shock that do not require hospitalization or long-term outpatient treatment.
2) Damage to property indicates extensive damage related to the destruction of assets or materials.

| NOTICE |
|---|
| • In order to install and dispose of the robot safety, be sure to read the “Safety Manual” Instruction Manual provided separately thoroughly before actually starting work. |

## Service Network

Please refer to our distributor's handouts.

# Specifications

This part describes the basic specifications, name of each part, coordinate system, and external dimensions of the dustproof and dripproof robot.

# 1. Specifications

## 1.1 Name of Each Part

Fig. 1.1 shows the name of each part of the robot.

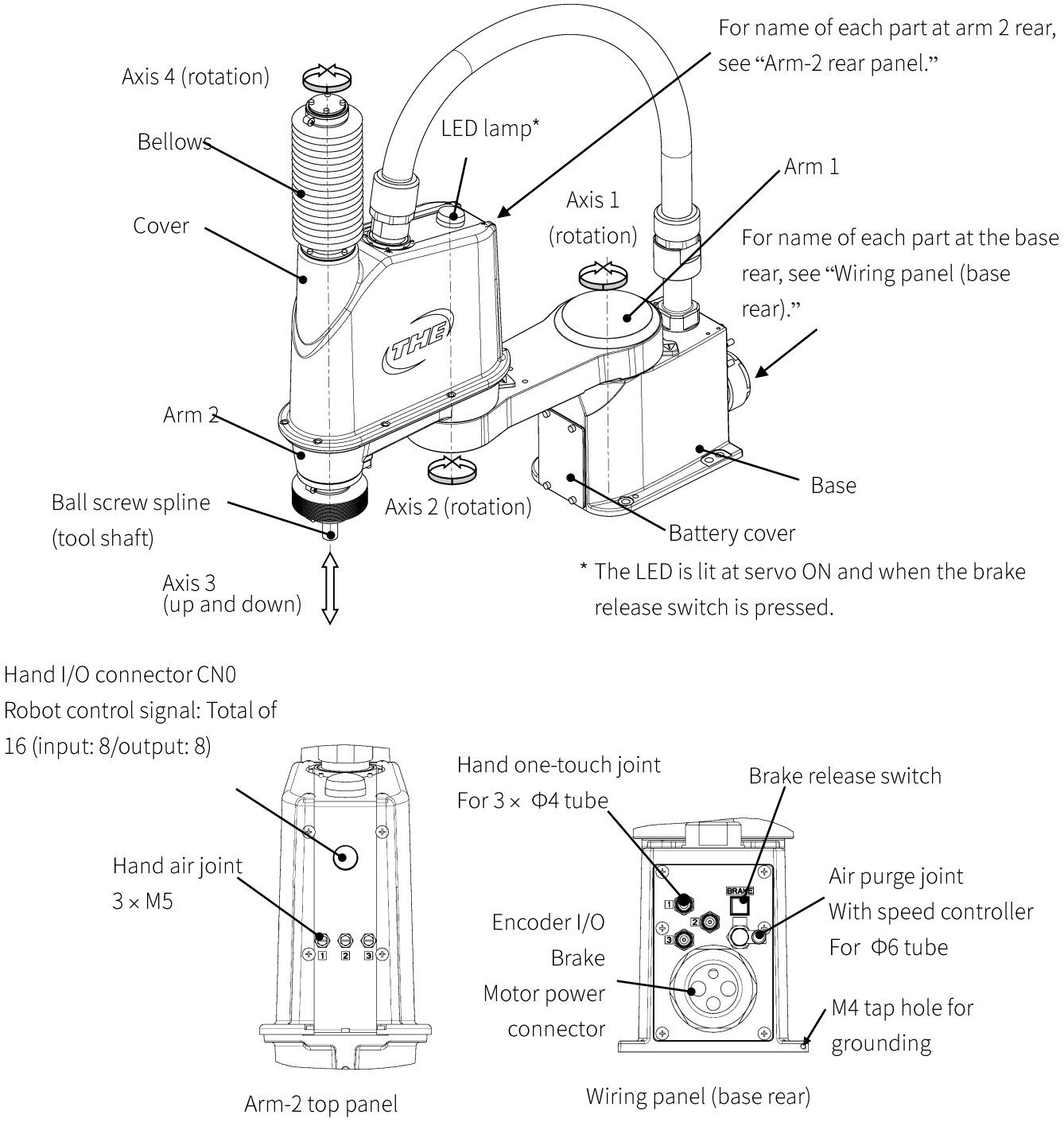



Fig. 1.1 Name of each part

## 1.2 Coordinate System

The robot's joint angle or stroke origin (0° or 0 mm position) is factory-calibrated according to the reference planes. Fig. 1.2 shows the base coordinate system and origin of each axis joint angle.

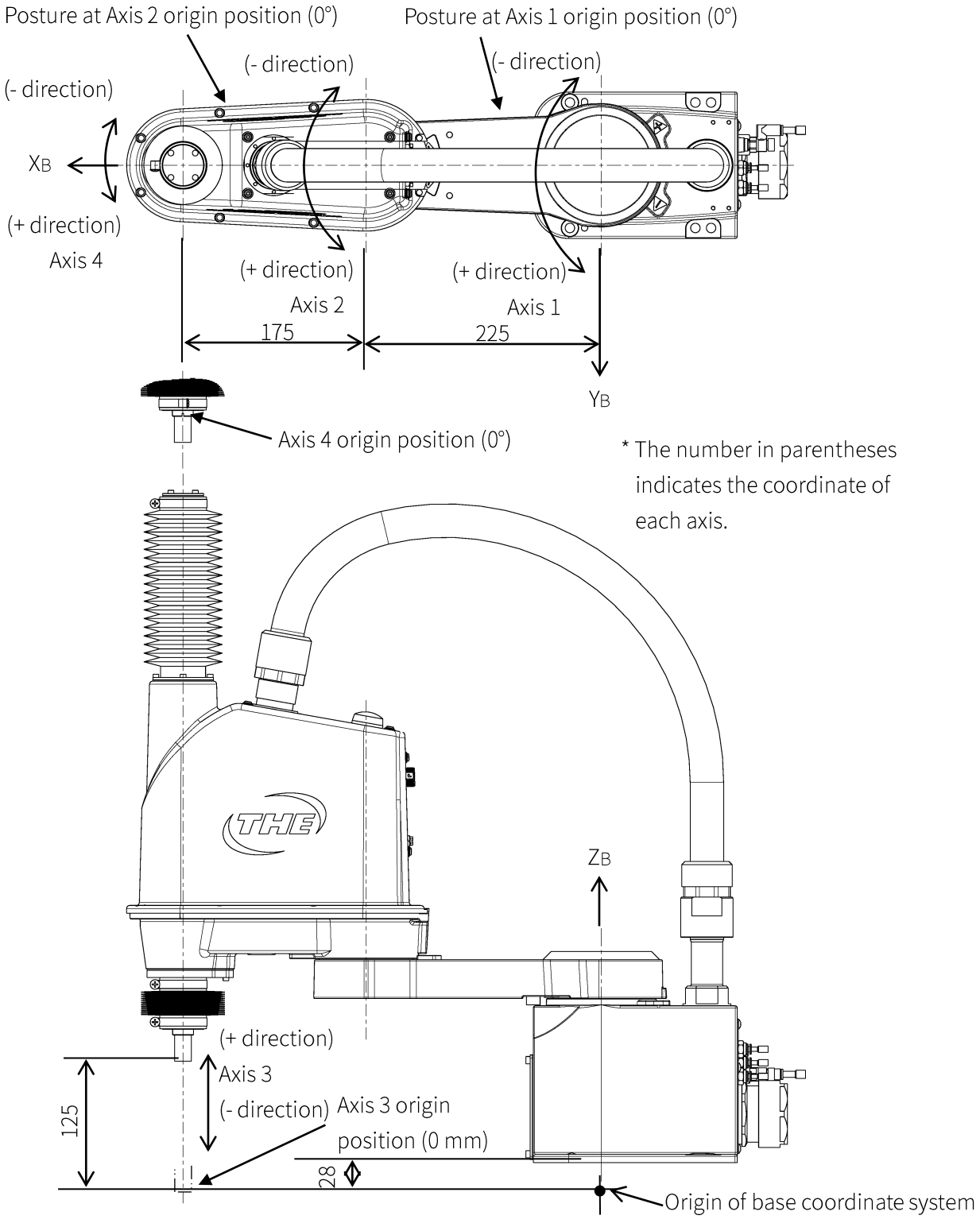



Fig. 1.2 Base coordinate system and joint angle origin

## 1.3 External Dimensions

Fig. 1.3 refers to the external dimensions of the Robot.

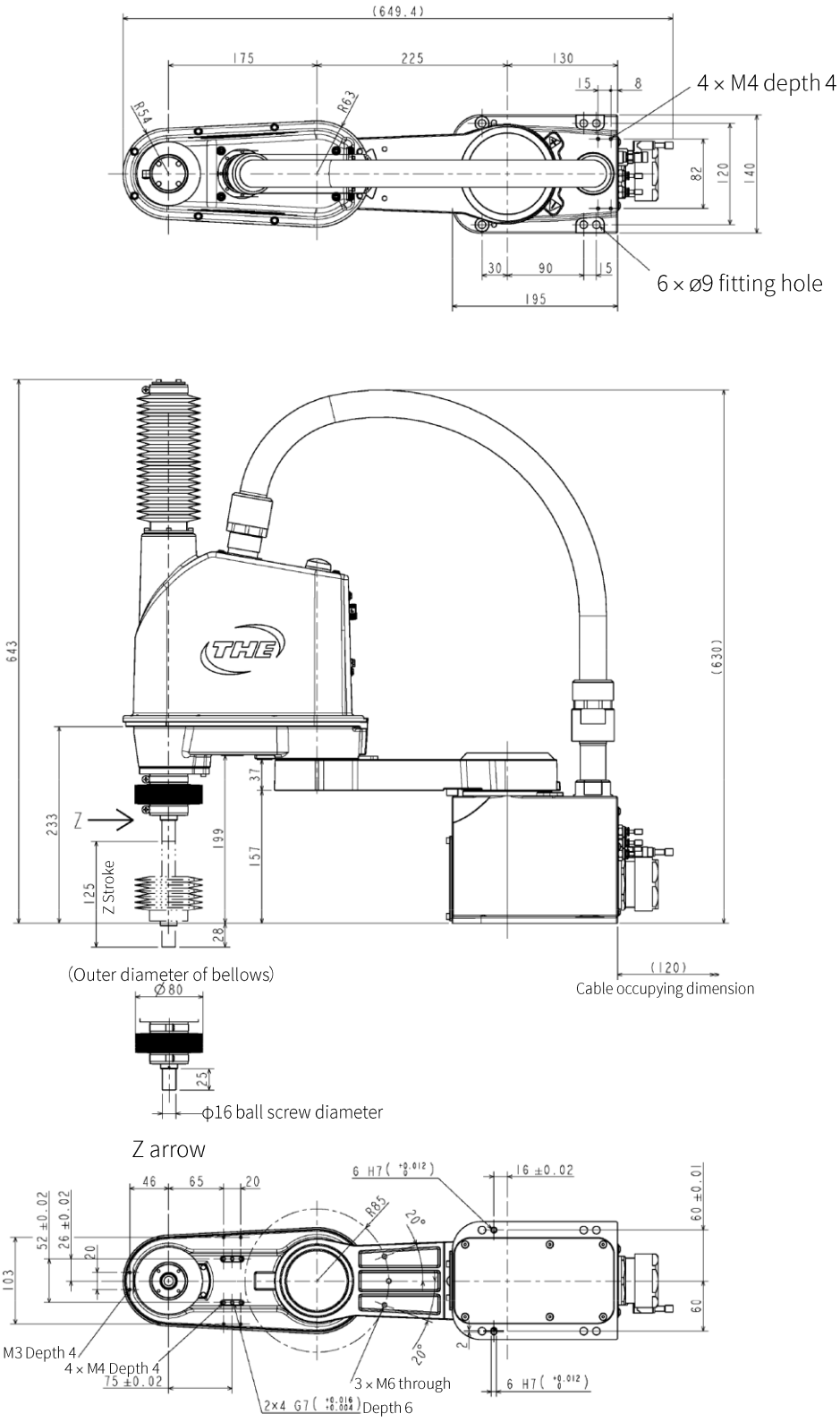



Fig. 1.3 External dimensions of the robot (THE400-IP)

## 1.4 Specifications Table

| Item | | Specifications |
|---|---|---|
| Structure | | Horizontally articulated type SCARA robot |
| Model | | THE400-IP |
| Applicable controller | | TSL3000, TSL3000E |
| Mass of robot body | | 15 kg |
| Number of controlled axes | | Four (4) |
| Arm length | | 400 mm (175 mm + 225 mm) |
| Motor capacity/current limit | Axis 1 | 400 W / 20.3 Ao-p |
| | Axis 2 | 200 W / 11.7 Ao-p |
| | Axis 3 | 100 W / 6.60 Ao-p |
| | Axis 4 | 100 W / 6.1 Ao-p |
| Operating range | Axis 1 | ±130 deg |
| | Axis 2 | ±145 deg |
| | Axis 3 | 0 to 125 mm |
| | Axis 4 | ±360 deg |
| Maximum speed *1 | Axis 1 | 672 deg/s |
| | Axis 2 | 780 deg/s |
| | Axis 3 | 1120 mm/s |
| | Axis 4 | 1800 deg/s |
| | Composite speed of axes 1 and 2 | 7.0 m/s |
| Rated payload mass*1 | | 1 kg |
| Maximum payload mass*1 | | 5 kg |
| Permissible load inertia*1 | | 0.06 kg·m$^2$ |
| Maximum offset of load gravity center | | 100 mm |
| Repeatability*2 | Composite speed of axes 1 and 2 | ±0.01 mm |
| | Axis 3 | ±0.01 mm |
| | Axis 4 | ±0.007 deg |
| Cycle time (when payload mass is 2 kg)*3 | | 0.39 sec |
| Drive system | | AC servo motor |
| Position detection method | | Absolute |
| Wiring for hand | | Input: 8, output: 8 |
| Paint color*4<br>Material color*4 | | Arm 2: PANTONE 293C or equivalent<br>Base and arm 1: White AN-90 or equivalent<br>Arm cover: White AN-90 or equivalent |
| Noise level*5 | | 68 dB (A) |
| Power supply capacity*6 | | 2.6kVA |

*1: The speed and acceleration are limited in accordance with the operation pattern, the load mass, and the offset value.

*2: Repeatability is the repeat accuracy in one direction at a constant ambient temperature and machine temperature. It is not absolute positioning accuracy.
It may exceed the specification values depending on the operation pattern, load weight, or offset value.
As for X-Y and C, they represent values at the upper limit of Z. Trajectory accuracy is not guaranteed.

*3: Continuous operation with standard cycle operation pattern and exceeding the effective load ratio cannot be performed.
During shuttle for rough positioning in horizontal direction of 300 mm and vertical direction of 25 mm.

*4: The color may differ according to each production lot. Be aware that there is no problem in the very nature of the product quality.

*5: The measurement conditions are as follows.

| | |
|---|---|
| Operating condition: | Rated load, concurrent operation of axes 1 and 2, maximum speed, and maximum acceleration |
| Measurement position: | Height of 1.6 m from the robot installation surface at a distance of 1 m from four front, rear, right and left directions of the robot |

*6: The measurement conditions are as follows.

| | |
|---|---|
| Operating condition: | Rated load and 2 kg, concurrent operation of all axes, maximum speed, and maximum acceleration |
| Measuring instrument: | Power analyzer (POWER ANALYZER 3390) |

## NOTICE

- Never use the robot in a high-temperature environment.
  Otherwise, the performance and lifespan of the alkaline battery used for motor position detector battery will be reduced.
  - If using the robot in a high-temperature environment, please consult with SHIBAURA MACHINE sales office.
- Never allow the lowered ball screw spline shaft to collide with obstacles etc.
  Otherwise, the ball screw spline shaft will be damaged before an alarm is generated.
- When micro vibration occurs, be sure to decrease the speed and decelerate the robot.
  Otherwise, the performance and lifespan of the robot will be reduced. The robot will become faulty.
  - Use the ACCEL/DECEL command to adjust the speed to the acceleration/deceleration at which the robot does not vibrate.
  - Micro vibration may occur on the robot tip depending on the posture of the robot.
- When operating Axis 1, 2, or 4, be sure to raise the ball screw spline shaft as much as possible.
  If the ball screw spline shaft is lowered, the shaft will be damaged early.
  - When operating Axis 1, 2, or 4 with the ball screw spline shaft lowered, use the SPEED command or ACCEL/DECEL command to adjust the operating speed or acceleration/deceleration to prevent the ball screw spline from vibrating.

# Installation

This part describes information you should know before unpacking, transporting, installing, and operating the robot.

## 2. Unpacking and Transportation

The worker should receive safety training defined by each country's laws and regulations. Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

| Type and name | Protection part and use | Recommended example |
|---|---|---|
| Helmet | Protection part: Head<br>Use: Protect from a falling object.<br>Protect from collision with the arm. |  |
| Safety glasses | Protection part: Eyes<br>Use: Protect from a flying object.<br>Protect from collision with the arm. |  |
| Protective gloves | Protection part: Hands and fingers<br>Use: Protect hands and fingers when caught in the machine.<br>Prevent a carried object from dropping. |  |
| Protective shoes | Protection part: Feet and toes<br>Use: Protect from a falling object. |  |

**⚠ CAUTION**

- Be sure to convey the product by two or more workers.
  Working in an unnatural posture will cause the packing box to drop.
  - Secure the transportation route in advance.
  - Use a cart etc. to carry the product.
  - Wear protective shoes.

**NOTICE**

- Be sure to convey the product by two or more workers.
  Working in an unnatural posture will cause the packing box to drop, resulting in a damage to the product in the box.
  - Do not use the dropped product.

## 2.1 Unpacking

The robot is shipped in corrugated cardboards. Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot.

After opening the packages, make sure that nothing has been damaged during transportation and that all accessories are included.

The packing posture and packed contents are the same as those for the standard machine. Refer to the Instruction Manual "STA1105 THE400 Robot Manual" provided separately.

For the dustproof and dripproof robot, observe the following precautions.

**NOTICE**

- Remove the vinyl bag carefully.
  Pulling it strongly will damage the bellows or covers.

## 2.2 Transportation

Move the robot very carefully. Make sure that no excessive impact or vibration is exerted on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all the clamp and base set bolts completely. When transporting the product to a separate location, secure it with the band as it was packed initially, and then put the product back into corrugated cardboards.

## 2.3 Mass and Dimensions

Fig. 2.1 shows the mass and external dimensions of the robot.

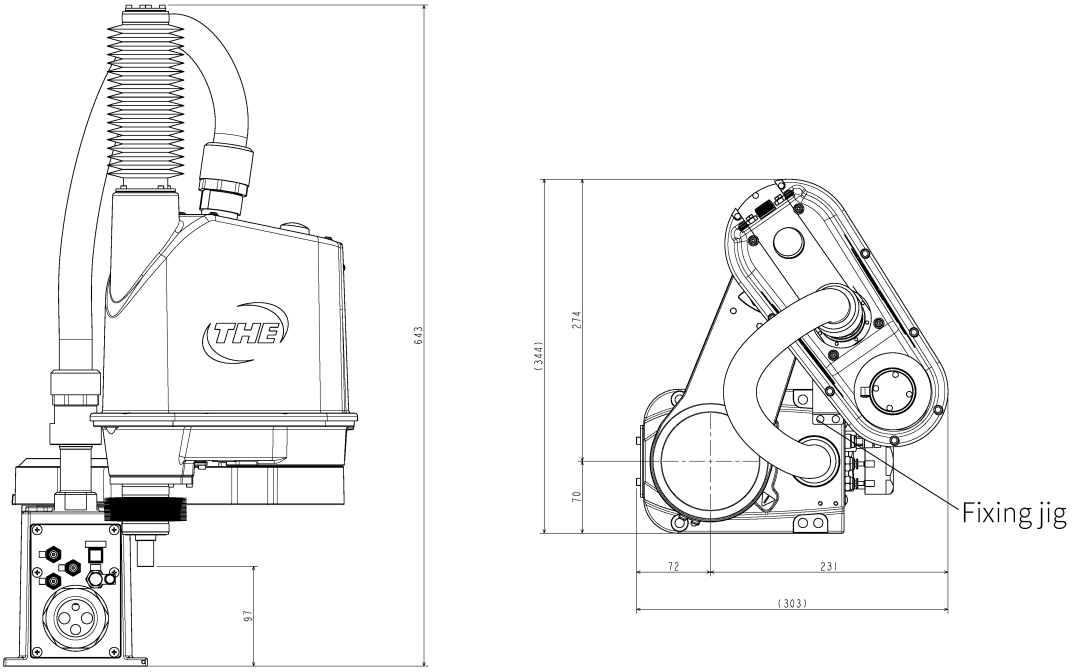



Fig. 2.1 Outer dimensions at transport (THE400-IP)

## 2.4 Transporting the Robot

The dustproof and dripproof robot has dustproof and dripproof parts such as bellows added to the standard machine.
The precautions for conveyance are shown below. The precautions other than the following are the same as those for the standard machine. Refer to the Instruction Manual “STA1105 THE400 Robot Manual” provided separately.

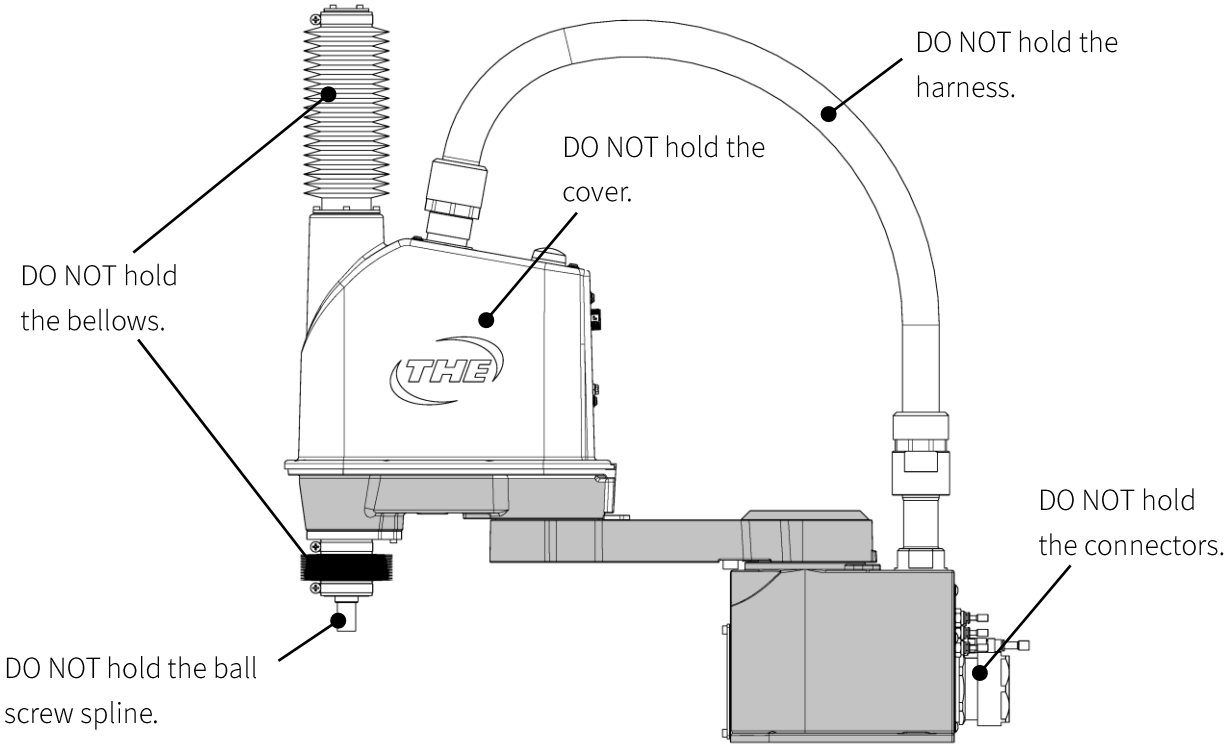



Fig. 2.2 Areas where holding is prohibited for the robot

**NOTICE**

- When lifting the robot by workers, be sure to hold the gray portions shown in Fig. 2.2 by hands.
  If other portions are held, the held portion will be damaged. Excessive force will be exerted on the mechanical section, resulting in a fault.
- Do not hold or grab the bellows.
  Otherwise, the bellows will be broken due to friction with internal parts.
- After installation is completed, be sure to remove the clamps.
  If the robot is moved with them mounted, the robot will be damaged or become faulty.

## 2.5 Storage

Storage of the robot is the same as that for the standard machine. For details, refer to the Instruction Manual “STA1105 THE400 Robot Manual”.

# 3. Installation of the Robot

The worker should receive safety training defined by each country's laws and regulations.
Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

| Type and name | Protection part and use | Recommended example |
|---|---|---|
| Helmet | Protection part: Head<br>Use: Protect from a falling object.<br>Protect from collision with the arm. |  |
| Safety glasses | Protection part: Eyes<br>Use: Protect from a flying object.<br>Protect from collision with the arm. |  |
| Protective gloves | Protection part: Hands and fingers<br>Use: Protect hands and fingers when caught in the machine.<br>Prevent a carried object from dropping. |  |
| Protective shoes | Protection part: Feet and toes<br>Use: Protect from a falling object. |  |

## 3.1 Installation Environment

Table 3.1 shows the environmental conditions for the location in which the robot are to be installed.

Table 3.1 Environmental conditions for robot

| Item | Specifications |
|---|---|
| Temperature | In operation: 0 to 40°C<br>In storage: -10°C to 50°C |
| Humidity | 20 to 80% (Non-condensing) |
| Altitude | 1000 m or less |
| Vibration | In operation: 0.98 $m/s^2$ or less |
| Dust | No inductive dust should exist. |
| Gas | No corrosive or combustible gas should exist. |
| IP (Ingress Protection) rating | IEC60529 IP65 |
| Overvoltage category | IEC60664-1 Class III |
| Protection against electric shock | IEC61140 Class I |
| Pollution degree | IEC60664-1 Pollution Degree 3 |
| Sunlight | The robot and controller should not be exposed to direct sunlight. |

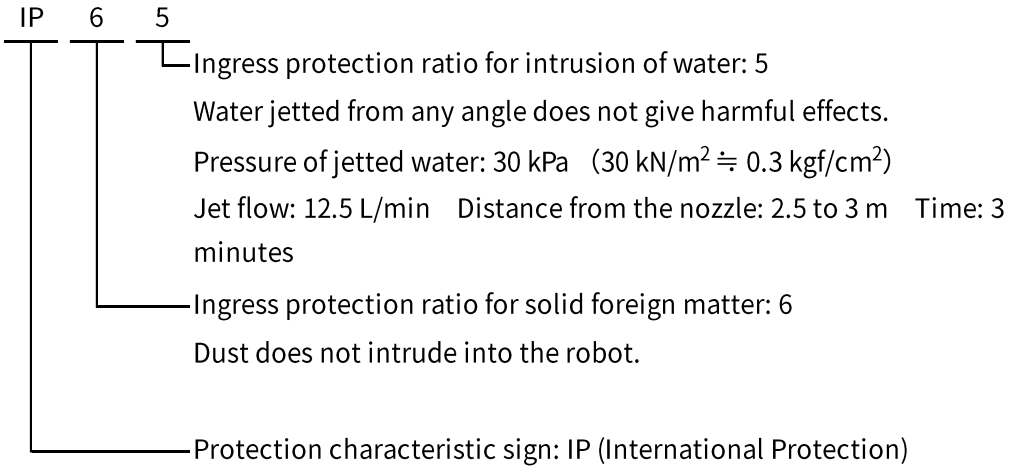
| Item | Specifications |
|---|---|
| Power noise | A heavy noise source should not exist nearby. |
| Magnetic field | A heavy magnetic field source should not exist nearby. |
| Other ambient environmental requirements | No iron powder, no oil, no salt content, and no organic solvent. |

**⚠ DANGER**

- Never use the robot at a place where water exceeding the Ingress Protection ratio or liquid other than water splashes.
  Otherwise, electric leakage will result in electric shock. The electric circuit will be short-circuited to cause fire.
- Never place a flammable material near the robot.
  If the robot ignites due to fault, fire will result.

## 3.2 Ingress Protection (IP) Rating for Dustproof and Dripproof Specification

The ingress protection rating of THE400 dustproof and dripproof specification is equivalent to IP65.
Be sure to purge air for use. Water or dust may enter the robot depending on the use environment.



## NOTICE

- Do not submerge the robot body or part of the body in water.
  Otherwise, water will intrude into the robot, and the robot will become faulty.
- Do not use the robot in the environment that exceeds the ingress protection ratio for water and dust.
  Otherwise, water or dust will intrude into the robot and the lifespan and operation accuracy of the robot will be reduced.
- Do not place the robot controller at a place that is exposed to water or dust.
  Otherwise, since the robot controller is neither dustproof nor dripproof, it will become faulty.
- Never use the robot at a place where water exceeding the Ingress Protection ratio or liquid other than water splashes.
  Otherwise, water etc. will intrude into the robot and the robot will become faulty.
  - For dripproof property for other than water, contact SHIBAURA MACHINE sales office.
- Be sure to purge air.
  If air is not purged, dustproof and dripproof performance will be reduced.

## 3.3 Air Purge

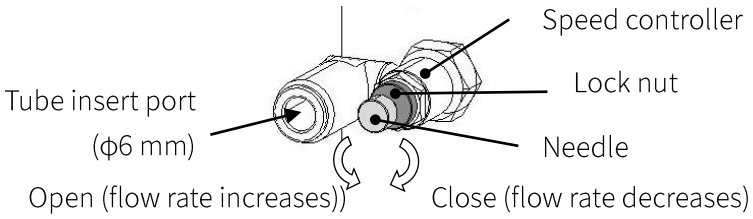
The dustproof and dripproof robot has an air supply port for air purge provided on the base connector.
See "Fig. 1.1 Name of each part." (One-touch joint with speed controller)
Air is supplied to the joint to prevent dust from intruding into the robot.
The customer should prepare the air source (pressure reduction valve, air filter, etc.) and air tube (ϕ6 mm).

### Air specifications

- Maximum operating pressure: 0.58 MPa (6 kgf/cm$^2$)
- Tube size: Outer diameter ϕ6 mm × inner diameter ϕ4 mm
- Fluid used: Clean dry air that does not contain compressor oil, etc.
  Air filter filterability: 10 μm or less

### Flow rate adjustment method

Fully close the speed controller attached to the body base connector.
Use the pressure reduction valve (prepared by the customer) to set the pressure from 0.1 MPa to 0.2 MPa (maximum pressure) and connect the air tube.
Turn the speed controller needle from fully closed state counterclockwise to increase the flow rate.
While checking the state of the bellows, adjust the flow rate and tighten the lock nut of the speed controller.
If too much air is supplied, the bellows will swell. Adjust the flow rate before the bellows swells.






NG OK

Fig. 3.1 Adjusting the bellows

**⚠ DANGER**

- Be sure to use clean dry air.
  If dry air is not used, condensation forms and water will accumulate inside the robot and electric leakage will result in electric shock.

**NOTICE**

- Do not apply pressure that is higher than the specified pressure.
  Otherwise, joint seals, bellows, etc. will be damaged and the dustproof and dripproof performance will be impaired.

## 3.4 Installation

Installation of the robot is the same as that for the standard machine. For details, refer to the Instruction Manual “STA1105 THE400 Robot Manual”.

# 4. Setting Hand

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations.
Be aware that failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

| Type and name | Protection part and use | Recommended example |
|---|---|---|
| Helmet | Protection part: Head<br>Use: Protect from a falling object.<br>Protect from collision with the arm. |  |
| Safety glasses | Protection part: Eyes<br>Use: Protect from a flying object.<br>Protect from collision with the arm. |  |
| Protective gloves | Protection part: Hands and fingers<br>Use: Protect hands and fingers when caught in the machine.<br>Prevent a carried object from dropping. |  |
| Protective shoes | Protection part: Feet and toes<br>Use: Protect from a falling object. |  |

## 4.1 Mounting hand

Fig. 4.1 shows the tip dimension of the ball screw spline on which the hand is mounted. As with the standard machine, the optional tool flange can also be used. For details on the tool flange, refer to the Instruction Manual "STA1105 THE400 Robot Manual". The hollow hole of the ball screw spline is blocked for dustproofing, and cannot be used for wiring etc.

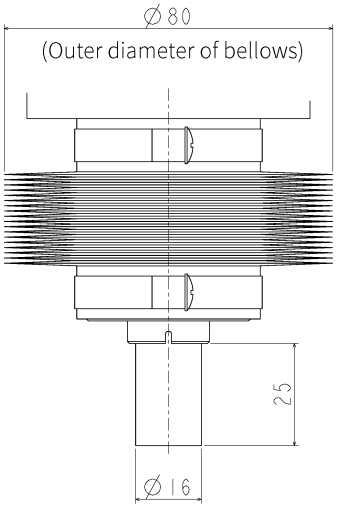



Fig. 4.1 Hand mounting dimensions

## 4.2 Mounting camera and air valve

Mounting of camera and air valve is the same as that for the standard machine. For details, refer to the "STA1105 THE400 Robot Manual."

## 4.3 Hand air piping

Three hand piping lines (Φ4 ×3) are available. The lines are connected on the base rear and the arm-2 rear.

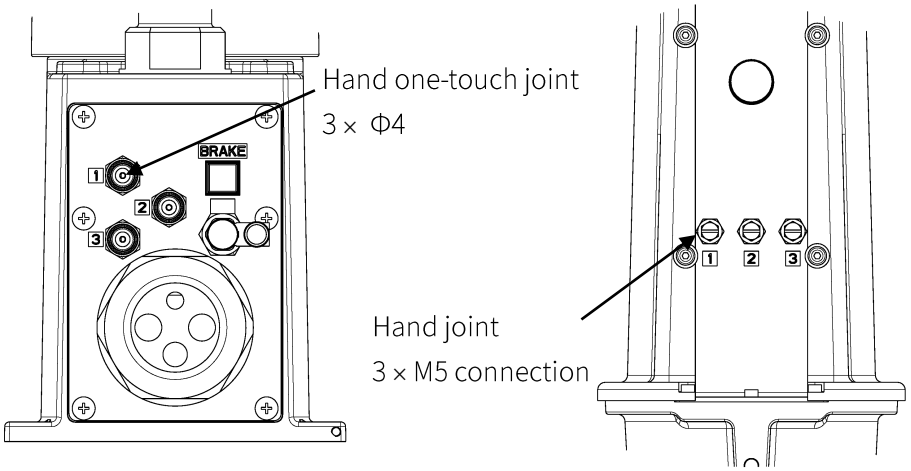



Fig. 4.2 Hand air piping and wiring

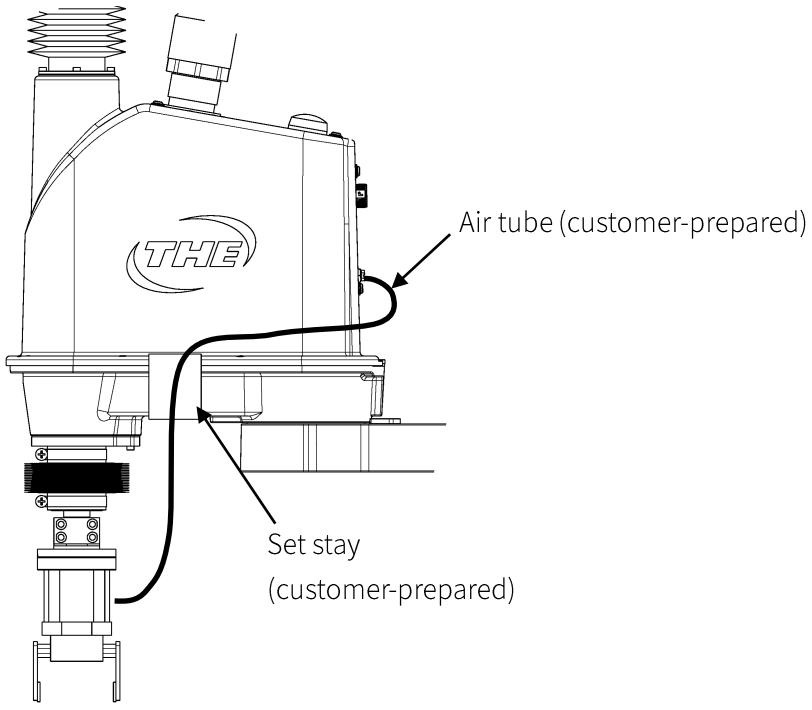



Fig. 4.3 Example of hand air piping and wiring

## 4.4 Permissible load conditions and program setting

Permissible load conditions and program setting are the same as those for the standard machine. For details, refer to the "STA1105 THE400 Robot Manual."

# 5. User Wiring

The work should receive safety training defined by each country's laws and regulations.
Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

## 5.1 Hand wiring

Hand wiring of the dustproof and dripproof robot is different from that for the standard machine.
This section describes the items that are different from those for the standard machine.

**⚠DANGER**

- When manufacturing a hand wiring cable, be sure to use an applicable wire.
  Otherwise, the cable will overheat or short circuit will occur between terminals to cause fire.
- Never connect the hand wiring cable with a power source whose voltage exceeds the specified range.
  Otherwise, the cable will overheat to cause fire.
  •Before connection, use a tester etc. to check the voltage of the power source.

# NOTICE

- Never mistake the terminal arrangement of the hand wiring.
  Otherwise, the connected parts will become faulty.
  - Before the use, use a tester etc. to check the terminal arrangement.
- When manufacturing a hand wiring cable, be sure to use an applicable wire.
  Otherwise, the cable will overheat or short circuit will occur between terminals to cause fire.
- Be sure to use a highly flexible and twisting-resistant robot cable for the hand wiring cable.
  If a general-purpose cable is used, the wire will be broken during robot operation.
  - Secure the cable to the arm with the saddle etc. to prevent the cable from being swung around.
- Never connect the hand wiring cable with a power source whose voltage exceeds the specified range.
  Otherwise, the cable will overheat to cause fire.
  - Before connection, use a tester etc. to check the voltage of the power source.
- Be sure to install hand wiring and piping while checking the following status during robot operation.
  Otherwise, the robot or hand will become faulty early.
  - Check whether a load is applied to the hand connector.
  - Check whether the hand wiring and piping contact the robot or peripheral equipment and are worn.

### 5.1.1 Hand control wiring

Eight input signals, eight output signals, and 24 VDC/PGND are provided as hand wiring and connected to the controller. Fig. 5.4 to Fig. 5.7 are the wiring diagrams. The wiring is connected to the round connector on the arm-2 rear.
The connector is not attached. The customer should prepare the connector and connect the cable.

Cannon connector type: <Body> LF13WBP-20P (manufacturer: HIROSE ELECTRIC)
Applicable cross sectional area of wire conductor: AWG 26 (plain braided wire and twisted wire)
The connector and wires are connected by soldering.
The signal line from the controller passes inside the robot and is wired to hand wiring connector "CN0."
Connect the manufactured cable connector to "CN0."

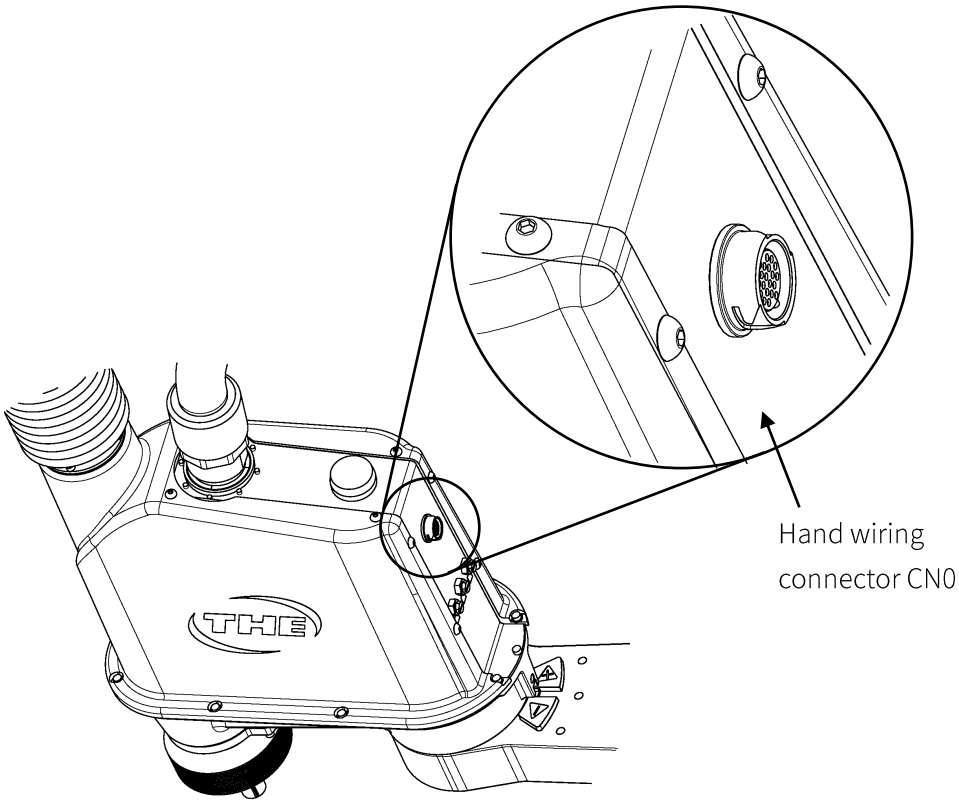



Fig. 5.1 Hand wiring

*MEMO*

*The hand I/O cable is also optionally available. For optional hand I/O cable, see "5.2 Hand I/O cable (option)."*

## 5.1.2 Specifications of hand connector CN0

Table 5.1 Input/output signal connector CN0 (TSL3000, Type-N)

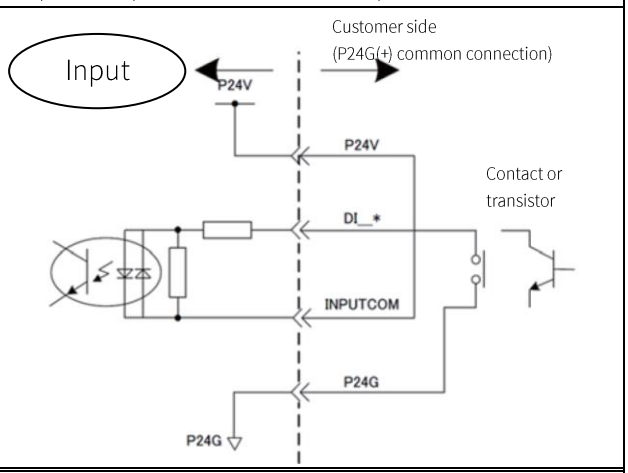
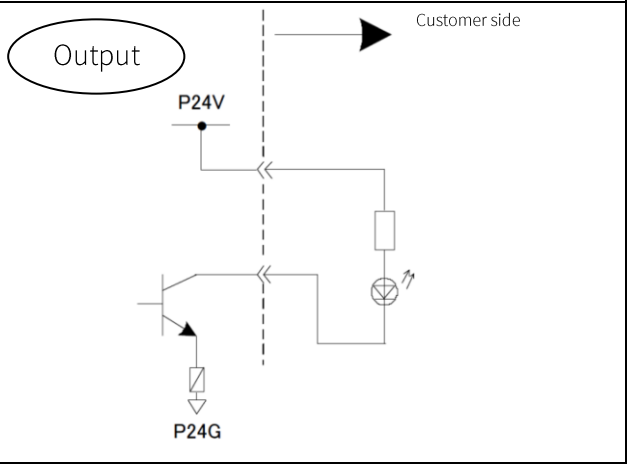
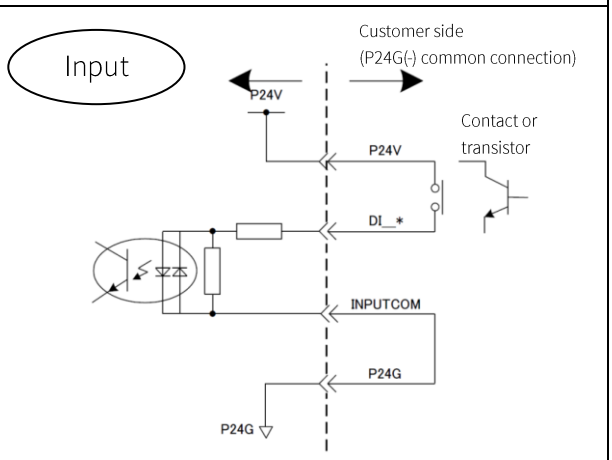
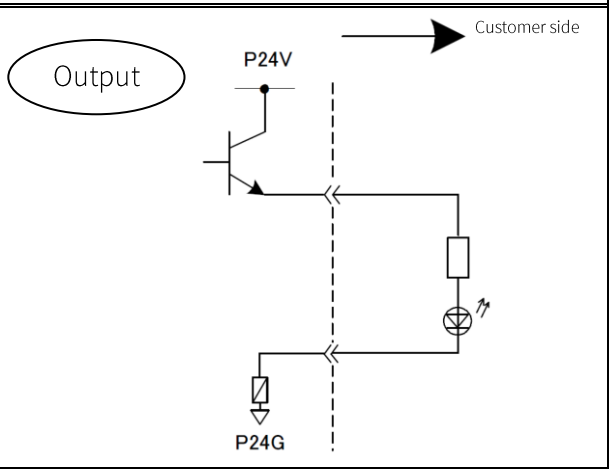
| Pin | Signal name | | Signal No. | Input/output circuit and example of connections |
|---|---|---|---|---|
| 1 | H-IN1 | Input signal 1 | 201 |  |
| 2 | H-IN2 | Input signal 2 | 202 | |
| 3 | H-IN3 | Input signal 3 | 203 | |
| 4 | H-IN4 | Input signal 4 | 204 | |
| 5 | H-IN5 | Input signal 5 | 205 | |
| 6 | H-IN6 | Input signal 6 | 206 | |
| 7 | H-IN7 | Input signal 7 | 207 | |
| 8 | H-IN8 | Input signal 8 | 208 | |
| 9 | P24G | | | |
| 10 | NC | | | |
| 11 | H-OUT1 | Output signal 1 | 201 |  |
| 12 | H-OUT2 | Output signal 2 | 202 | |
| 13 | H-OUT3 | Output signal 3 | 203 | |
| 14 | H-OUT4 | Output signal 4 | 204 | |
| 15 | H-OUT5 | Output signal 5 | 205 | |
| 16 | H-OUT6 | Output signal 6 | 206 | |
| 17 | H-OUT7 | Output signal 7 | 207 | |
| 18 | H-OUT8 | Output signal 8 | 208 | |
| 19 | P24V | | | |
| 20 | FG | | | |

Table 5.2 Input/output signal connector CN0 (TSL3000, Type-P)

| Pin | Signal name | | Signal No. | Input/output circuit and example of connections |
|---|---|---|---|---|
| 1 | H-IN1 | Input signal 1 | 201 |  |
| 2 | H-IN2 | Input signal 2 | 202 | |
| 3 | H-IN3 | Input signal 3 | 203 | |
| 4 | H-IN4 | Input signal 4 | 204 | |
| 5 | H-IN5 | Input signal 5 | 205 | |
| 6 | H-IN6 | Input signal 6 | 206 | |
| 7 | H-IN7 | Input signal 7 | 207 | |
| 8 | H-IN8 | Input signal 8 | 208 | |
| 9 | P24V | | | |
| 10 | NC | | | |
| 11 | H-OUT1 | Output signal 1 | 201 |  |
| 12 | H-OUT2 | Output signal 2 | 202 | |
| 13 | H-OUT3 | Output signal 3 | 203 | |
| 14 | H-OUT4 | Output signal 4 | 204 | |
| 15 | H-OUT5 | Output signal 5 | 205 | |
| 16 | H-OUT6 | Output signal 6 | 206 | |
| 17 | H-OUT7 | Output signal 7 | 207 | |
| 18 | H-OUT8 | Output signal 8 | 208 | |
| 19 | P24G | | | |
| 20 | FG | | | |





Input signal

As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

| | |
|---|---|
| Contact rating: | 24 VDC, 10 mA or over (circuit current: approx. 7 mA) |
| Minimum contact current: | 24 VDC, 1 mA |
| Contact impedance: | 100 Ω or less |

Transistor specifications:

| | |
|---|---|
| Withhold voltage between collector and emitter: | 30 V or over |
| Current between collector and emitter: | 10 mA or over (circuit current: approx. 7 mA) |
| Leak current between collector and emitter: | 100 μA or less |

Output signal

By using the P24V of the controller, a relay, solenoid valve, etc. can be driven.

To use external power supply, connect GND of the external power supply with the GND (P24G) of the robot controller (GND common connection).

| | | |
|---|---|---|
| Transistor specifications: | Rated voltage: | 24 VDC (MAX 30 V) |
| | Rated current: | 100 mA |

- Limit P24V supplied from the robot controller to within 3 A in total.

- To use external power supply, limit the total current value to within 3 A as well
- To connect a relay, solenoid valve, etc., use a surge killer or diode to absorb the surge voltage.

Table 5.3 Input/output signal connector CN0 (TSL3000E, Type-N)

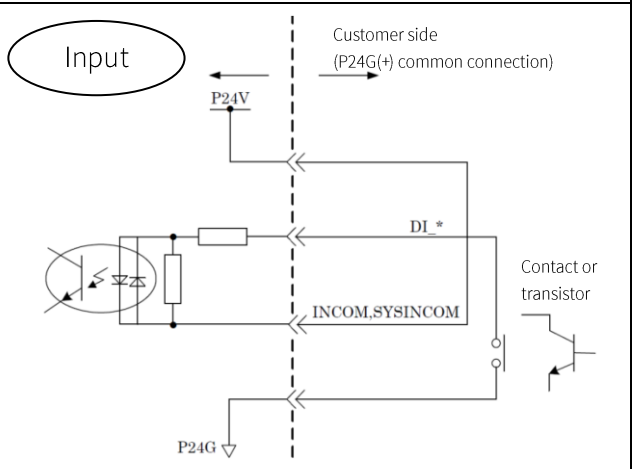
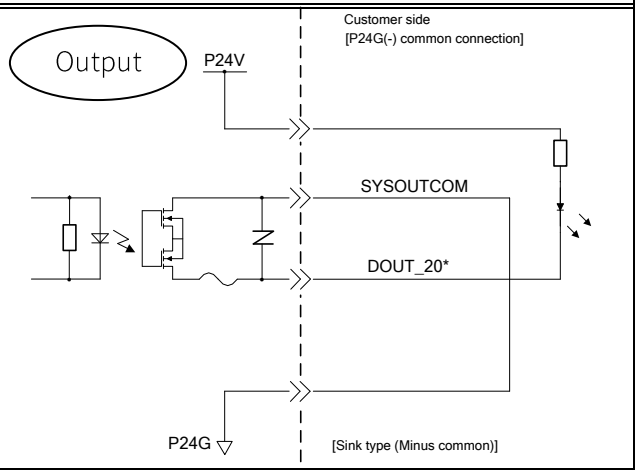
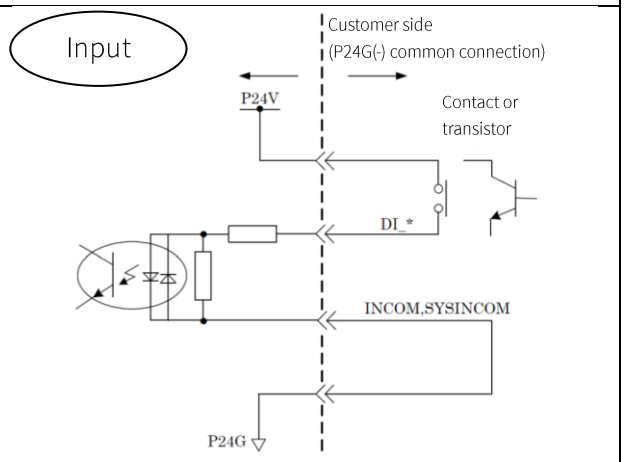
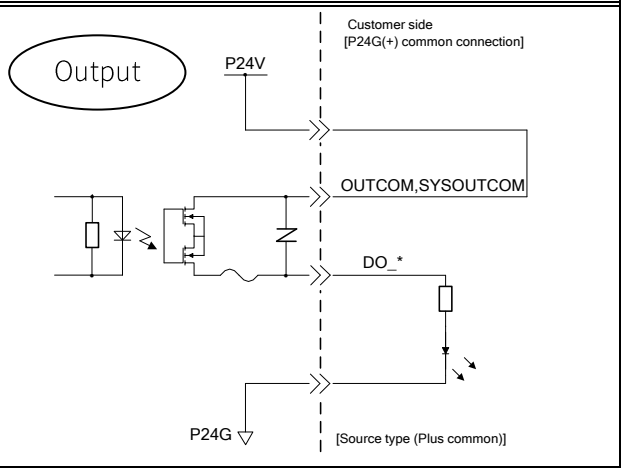
| Pin | Signal name | | Signal No. | Input/output circuit and example of connections |
|---|---|---|---|---|
| 1 | H-IN1 | Input signal 1 | 201 |  |
| 2 | H-IN2 | Input signal 2 | 202 | |
| 3 | H-IN3 | Input signal 3 | 203 | |
| 4 | H-IN4 | Input signal 4 | 204 | |
| 5 | H-IN5 | Input signal 5 | 205 | |
| 6 | H-IN6 | Input signal 6 | 206 | |
| 7 | H-IN7 | Input signal 7 | 207 | |
| 8 | H-IN8 | Input signal 8 | 208 | |
| 9 | P24V | | | |
| 10 | NC | | | |
| 11 | H-OUT1 | Output signal 1 | 201 |  |
| 12 | H-OUT2 | Output signal 2 | 202 | |
| 13 | H-OUT3 | Output signal 3 | 203 | |
| 14 | H-OUT4 | Output signal 4 | 204 | |
| 15 | H-OUT5 | Output signal 5 | 205 | |
| 16 | H-OUT6 | Output signal 6 | 206 | |
| 17 | H-OUT7 | Output signal 7 | 207 | |
| 18 | H-OUT8 | Output signal 8 | 208 | |
| 19 | P24G | | | |
| 20 | FG | | | |

Table 5.4 Input/output signal connector CN0 (TSL3000E, Type-P)

| Pin | Signal name | | Signal No. | Input/output circuit and example of connections |
|---|---|---|---|---|
| 1 | H-IN1 | Input signal 1 | 201 |  |
| 2 | H-IN2 | Input signal 2 | 202 | |
| 3 | H-IN3 | Input signal 3 | 203 | |
| 4 | H-IN4 | Input signal 4 | 204 | |
| 5 | H-IN5 | Input signal 5 | 205 | |
| 6 | H-IN6 | Input signal 6 | 206 | |
| 7 | H-IN7 | Input signal 7 | 207 | |
| 8 | H-IN8 | Input signal 8 | 208 | |
| 9 | P24V | | | |
| 10 | NC | | | |
| 11 | H-OUT1 | Output signal 1 | 201 |  |
| 12 | H-OUT2 | Output signal 2 | 202 | |
| 13 | H-OUT3 | Output signal 3 | 203 | |
| 14 | H-OUT4 | Output signal 4 | 204 | |
| 15 | H-OUT5 | Output signal 5 | 205 | |
| 16 | H-OUT6 | Output signal 6 | 206 | |
| 17 | H-OUT7 | Output signal 7 | 207 | |
| 18 | H-OUT8 | Output signal 8 | 208 | |
| 19 | P24G | | | |
| 20 | FG | | | |





Input signal

As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

| | |
|---|---|
| Contact rating: | 24 VDC, 10 mA or over (circuit current: approx. 7 mA) |
| Minimum contact current: | 24 VDC, 1 mA |
| Contact impedance: | 100 Ω or less |

Transistor specifications:

| | |
|---|---|
| Withhold voltage between collector and emitter: | 30 V or over |
| Current between collector and emitter: | 10 mA or over (circuit current: approx. 7 mA) |
| Leak current between collector and emitter: | 100 μA or less |

Output signal

By using P24V of the controller, a relay, solenoid valve, etc. can be driven.

To use the external power supply, connect GND of the external power supply with GND (P24G) of the robot controller (GND common connection).

Photo MOS specifications:

| | |
|---|---|
| Rated voltage: | 24 VDC (MAX 30 V) |
| Rated current: | 50mA (MAX) |

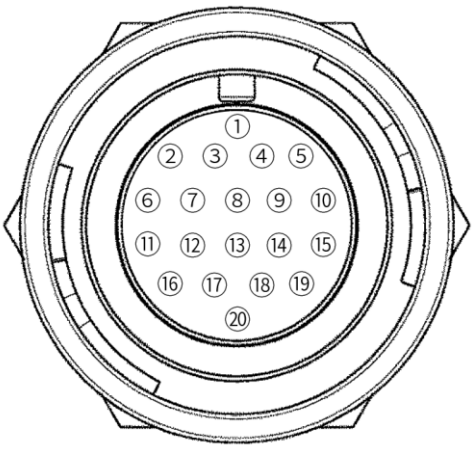



Fig. 5.2 Hand connector pin arrangement

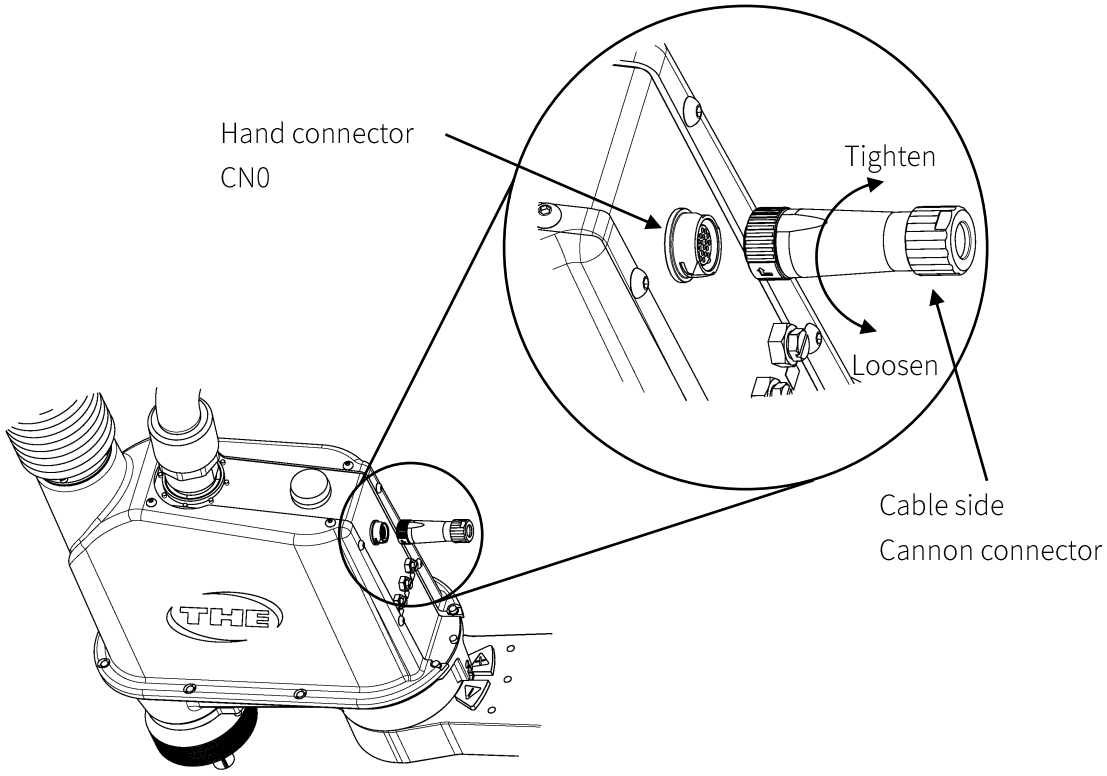



Fig. 5.3 Connecting/disconnecting hand input/output signal connector

## ⚠ CAUTION

- Be sure to connect the connectors securely.
  If the connectors are about to come off, the robot hand will move unpredictably.
  - Since the hand wiring cable is swung around during robot operation, secure the cable to the arm with the clamp, etc.
  - Adjust the robot wiring and piping while checking the status during operation to prevent them from contacting the robot and peripheral equipment and applying load to the connector.

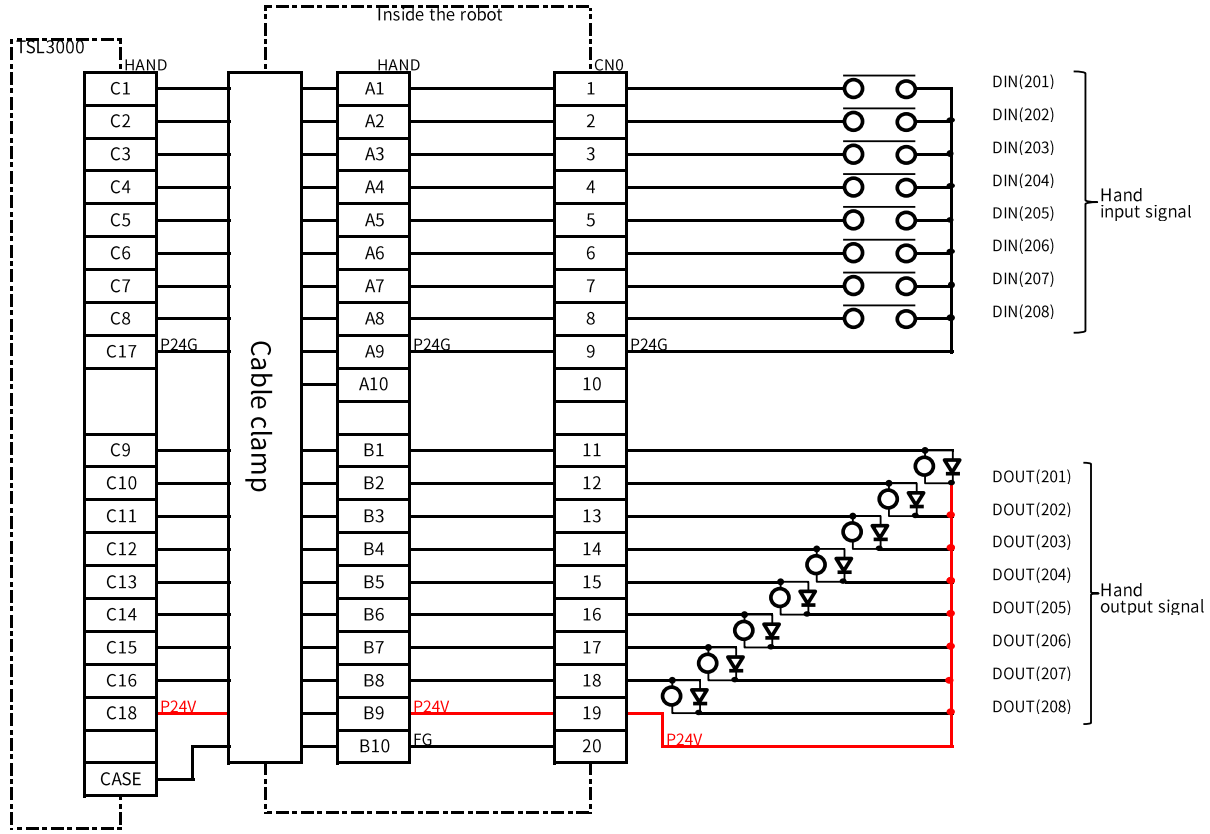



Fig. 5.4 Hand wiring diagram (Type-N, TSL3000)

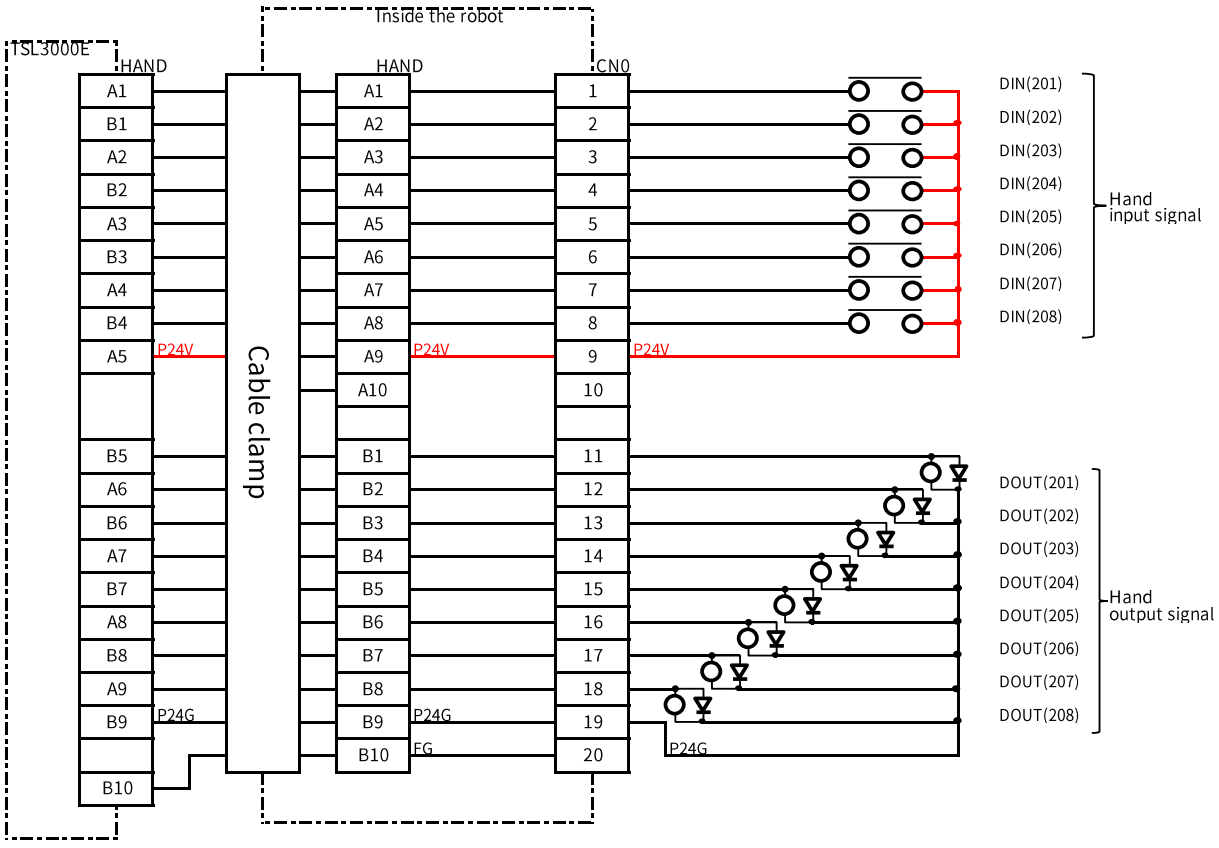



Fig. 5.5 Hand wiring diagram (Type-P, TSL3000)

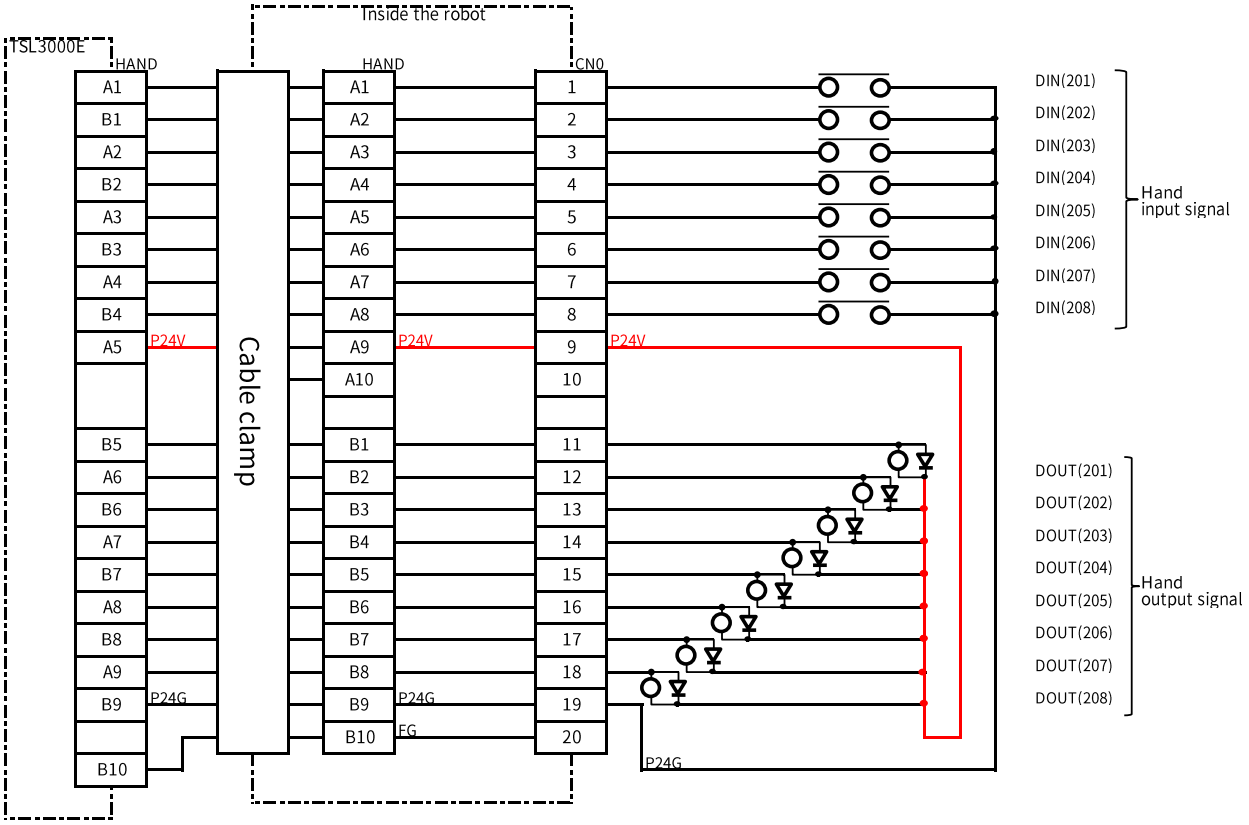



Fig. 5.6 Hand wiring diagram (Type-N, TSL3000E)

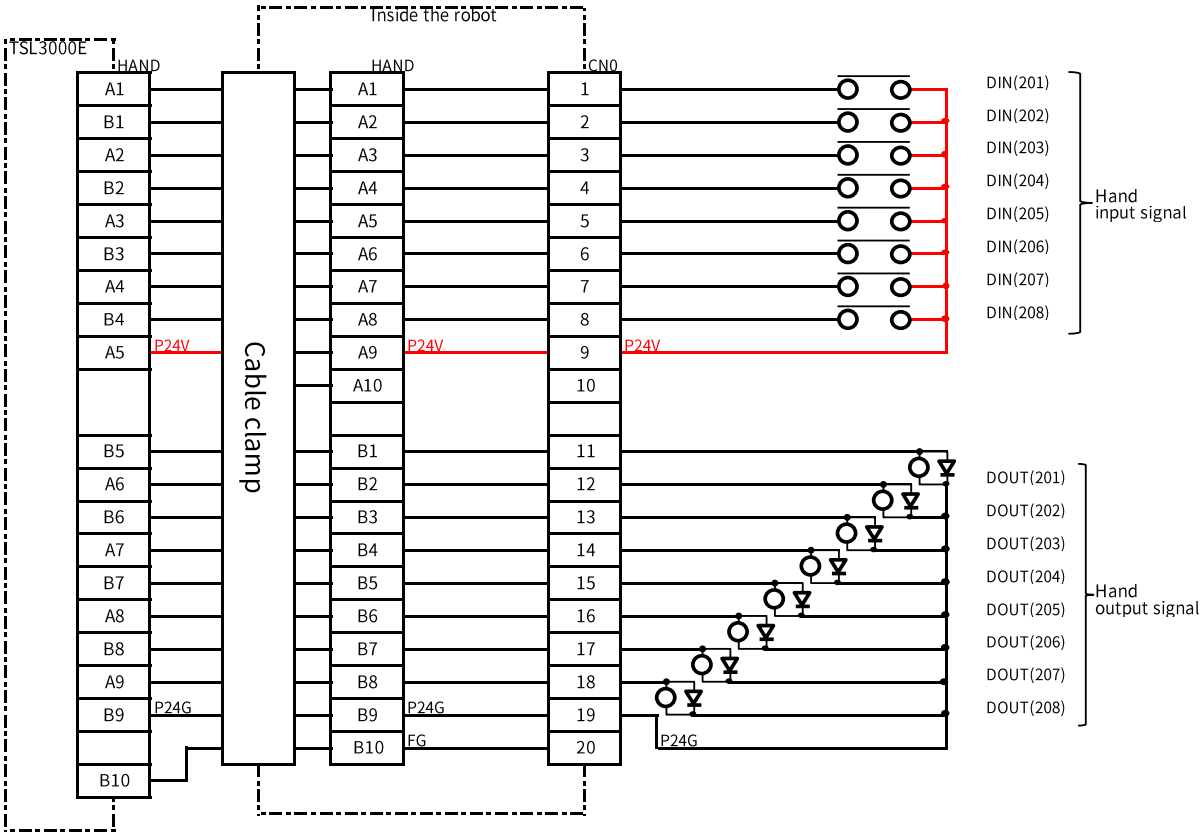



Fig. 5.7 Hand wiring diagram (Type-P, TSL3000E)

### 5.1.3 Connection example

Fig. 5.8 shows an example of hand wiring.

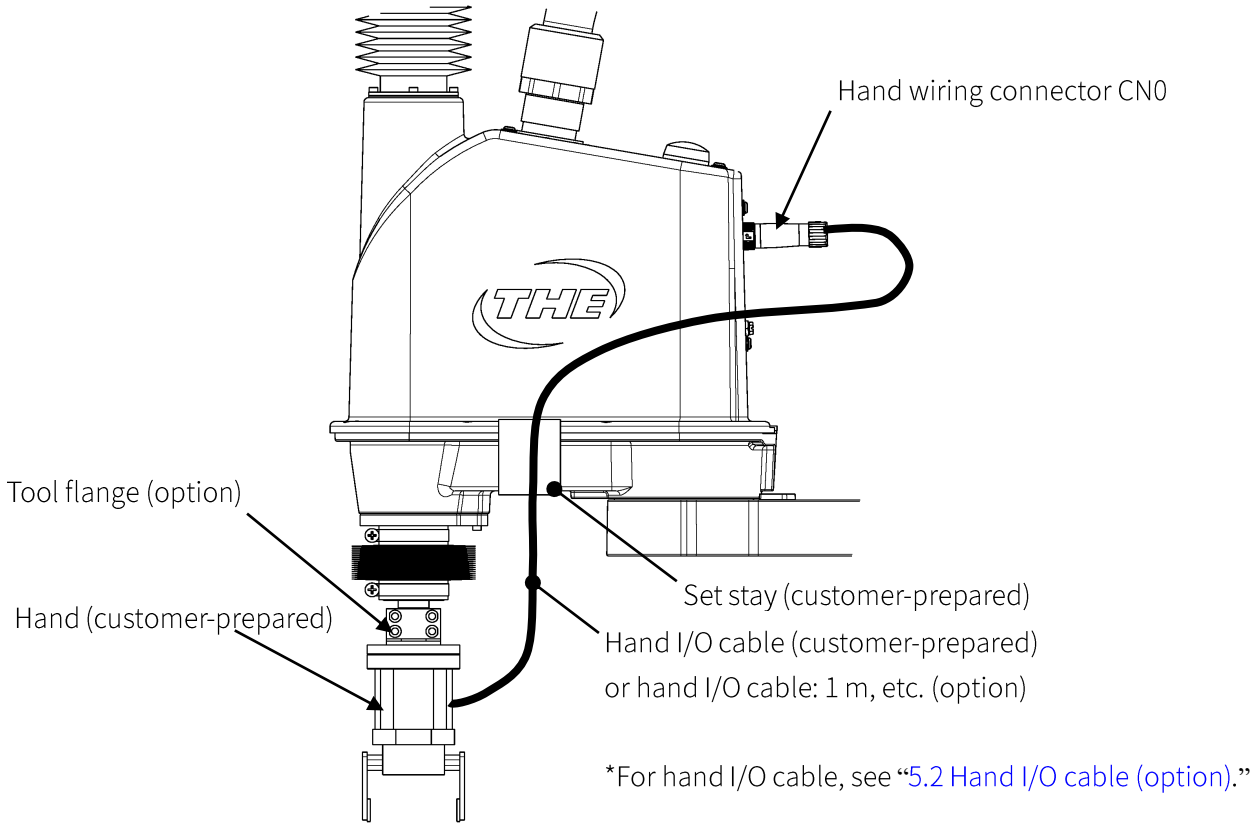



Fig. 5.8 Example of hand I/O cable wiring to arm-2 top

For the mounting dimensions of the set stay, see the "STA1105 THE400 Robot Manual."

**NOTICE**

- Be sure to perform hand wiring and piping while checking the following status during robot operation. Otherwise, the robot and hand will become faulty early.
  - Check whether a load is applied to the hand connector.
  - Check whether hand wiring and piping contact the robot and peripheral equipment and are worn.
  - Make sure that the securing stay does not interfere with arm 1 etc.

## 5.2 Hand I/O cable (option)

Fig. 5.9 and Fig. 5.10 are external view and wiring diagram of the hand I/O cable (option), respectively.

Table 5.5 Hand I/O cable (option) parts usedTable 5.5 Hand I/O cable (option) parts used lists parts used, and Table 5.6 Hand I/O cable (option) lists SHIBAURA MACHINE drawing number of the hand I/O cable (option) for reference.

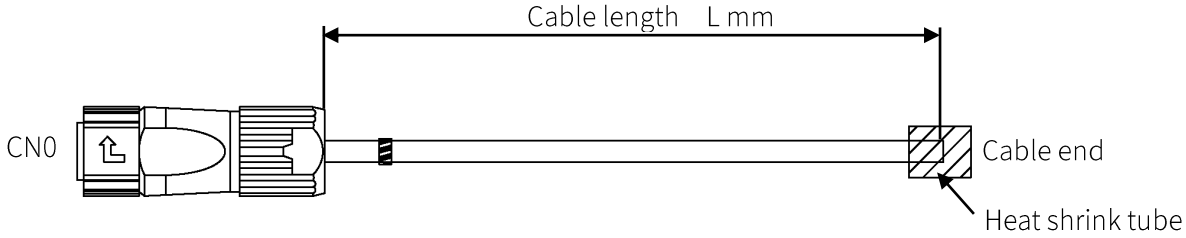



Fig. 5.9 External view drawing of hand I/O cable (option)

| CN0 | | | Cable end | |
|---|---|---|---|---|
| Pin | Signal name | Color | Signal name | Color |
| 1 | H-IN1 | Blue | H-IN1 | Blue |
| 2 | H-IN2 | White | H-IN2 | White |
| 3 | H-IN3 | Yellow | H-IN3 | Yellow |
| 4 | H-IN4 | White | H-IN4 | White |
| 5 | H-IN5 | Green | H-IN5 | Green |
| 6 | H-IN6 | White | H-IN6 | White |
| 7 | H-IN7 | Red | H-IN7 | Red |
| 8 | H-IN8 | White | H-IN8 | White |
| 9 | INPUT COM | Purple | INPUT COM | Purple |
| 10 | NC | | NC | |
| 11 | H-OUT1 | Blue | H-OUT1 | Blue |
| 12 | H-OUT2 | Brown | H-OUT2 | Brown |
| 13 | H-OUT3 | Yellow | H-OUT3 | Yellow |
| 14 | H-OUT4 | Brown | H-OUT4 | Brown |
| 15 | H-OUT5 | Green | H-OUT5 | Green |
| 16 | H-OUT6 | Brown | H-OUT6 | Brown |
| 17 | H-OUT7 | Red | H-OUT7 | Red |
| 18 | H-OUT8 | Brown | H-OUT8 | Brown |
| 19 | OUTPUT COM | Purple | OUTPUT COM | Purple |
| 20 | FG | | FG | |

Fig. 5.10 Hand I/O cable (option) wiring diagram

Table 5.5 Hand I/O cable (option) parts used

| No. | Product name | Type | Manufacturer | Use qt. |
|---|---|---|---|---|
| 1 | Cable | RMFEV-SB(CL3) 0.2SQ-10P | DYDEN Corp. | 1 |
| 2 | Cannon connector | LF13WBP-20P | HIROSE ELECTRIC CO., LTD. | 1 |

Table 5.6 Hand I/O cable (option)

| Product name | Type | Cable length | Shibaura Dwg. No. | Unit code |
|---|---|---|---|---|
| Hand I/O cable | | 1m | 395N7001 | Y610D08Q0 |
| | | 1.5m | 395N7002 | Y610D08R0 |
| | | 2m | 395N7003 | Y610D08S0 |
| | | 2.5m | 395N7004 | Y610D08T0 |
| | | 3m | 395N7005 | Y610D08U0 |

# Maintenance Manual

This manual describes how to maintain respective units of the THE series robot.

# 6. Maintenance

This section describes the maintenance and inspection items for the dustproof and dripproof robot. For the inspection detail, refer to the Instruction Manual "STA1105 THE400 Robot Manual."

## 6.1 Maintenance Schedule

Table 6.1 shows the maintenance guidelines for the dustproof and dripproof robot. Other maintenance items are the same as those for the standard machine. Refer to the Instruction Manual “STA1105 THE400 Robot Manual”.

Table 6.1 Maintenance Items for Dustproof and Dripproof Robot

| Description | Inspection guideline | Daily check | Every 3 months | Every 6 months | Every year | Refer to |
|---|---|---|---|---|---|---|
| Check of bellows | Visually check whether upper and lower bellows are damaged, cracked, twisted, etc. | ○ | ○ | ○ | ○ | 7.1.1 |
| Replacement of bellows | Replace the bellows if they are damaged etc. or according to the schedule. | | | | ○ | 7.1.2<br>7.3 |
| Greasing of ball screw spline unit | Remove the bellows, grease the ball screw spline, and mount the bellows. | | ○ | ○ | ○ | 7.1.3<br>7.3.1<br>STA1105 |

**NOTICE**

- Intrusion of dust into the robot will cause a fault. Perform maintenance work such as removal of bellows in a dust-free environment.

## 6.2 Maintenance Tools

As the maintenance tools and provisions, it is recommended to prepare the following item.
For tools and provisions other than the following item, refer to the Instruction Manual “STA1105 THE400 Robot Manual”.

- Liquid gasket (recommended product: 1221H manufacturer: ThreeBond)

# 7. Maintenance of the Robot Body

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations. Shibaura Machine Service Department also accepts replacement work at an extra cost.

Be aware that failure and accident resulting from the work done by the customer will not be under our warranty.

The basic maintenance contents of the dustproof and dripproof robot are the same as those for the standard machine.

Refer to the Instruction Manual "STA1105 THE400 Robot Manual".

The items that are different from those for the standard machine are described below.

Recommended protector:

| Type and name | Protection part and use | Recommended example |
|---|---|---|
| Helmet | Protection part: Head<br>Use: Protect from a falling object.<br>Protect from collision with the arm. |  |
| Safety glasses | Protection part: Eyes<br>Use: Protect from a flying object.<br>Protect from collision with the arm. |  |
| Protective gloves | Protection part: Hands and fingers<br>Use: Protect hands and fingers when caught in the machine.<br>Prevent a carried object from dropping. |  |
| Protective shoes | Protection part: Feet and toes<br>Use: Protect from a falling object. |  |

## 7.1 Details of Inspection

### 7.1.1 Check of bellows

While the robot is operating or stopped, visually check that breakage due to peeling, scratch, or wear and abnormalities such as twist and wrinkle have not occurred in the bellows. Since breakage will impair the dustproof and dripproof performance to cause a fault in the robot, replace the bellows immediately if there is a breakage in the bellows. Since abnormalities such as twist and wrinkle will cause breakage, adjust the flow rate of air purge and remount the bellows if any abnormality is found in them.

### 7.1.2 Replacement of bellows

Replace the bellows if there is breakage in the bellows or according to the schedule.
Replace the bellows in 30 million strokes of axis 3 or one year, whichever is reached earlier, as the schedule.

### 7.1.3 Greasing of ball screw spline unit

See the "Bellows Replacement Procedure" to remove the bellows, and refer to the Instruction Manual "STA1105 THE400 Robot Manual" to grease up the ball screw spline unit.

## 7.2 Configuration of Mechanical Components

Fig. 7.1 shows the configuration of the mechanical components.

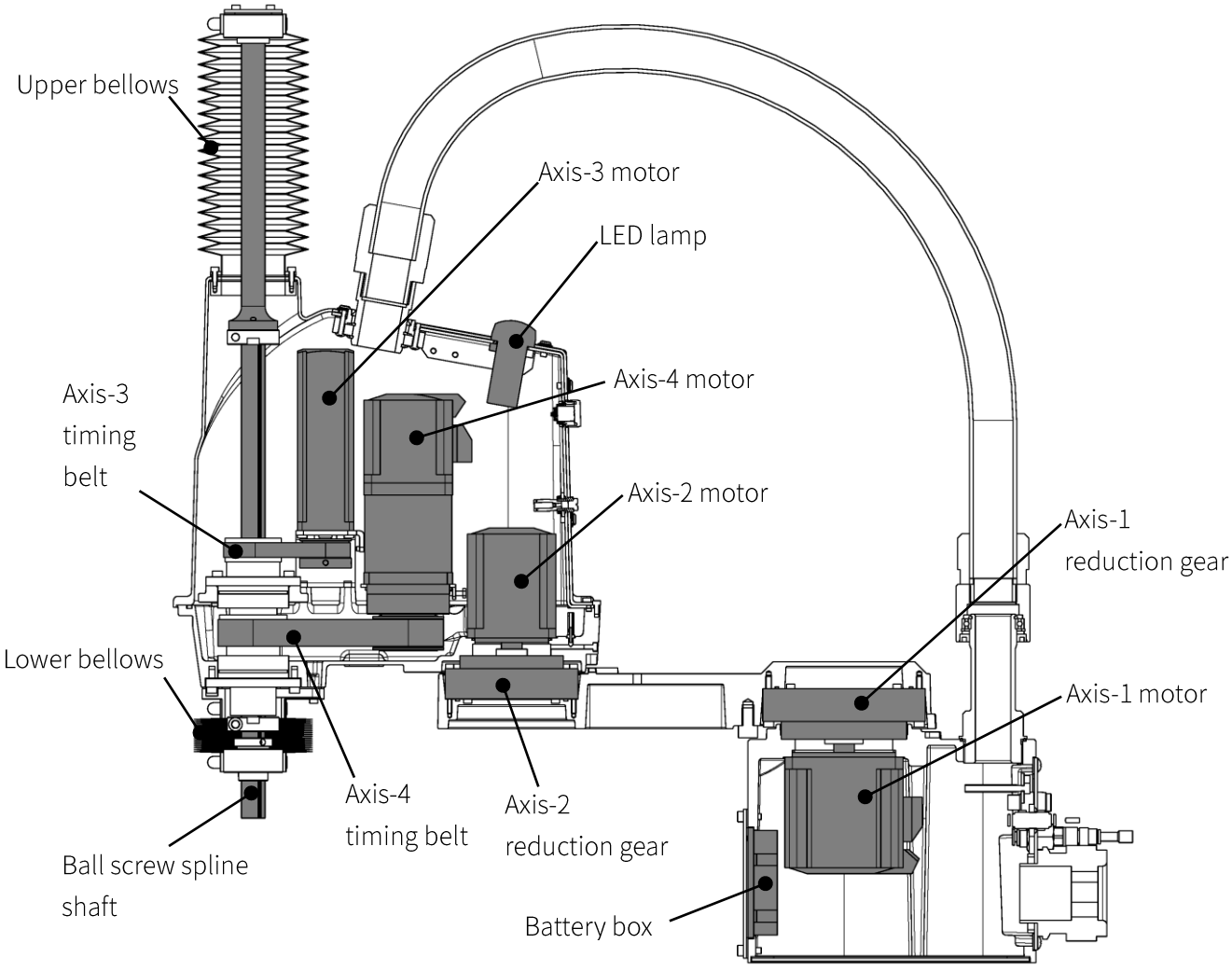



Fig. 7.1 Configuration of mechanical components

## 7.3 Bellows Replacement Procedure

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations. SHIBAURA MACHINE Service Department also accepts replacement work at an extra cost.

Be aware that the fault and accident resulting from the replacement of parts by the customer will not be under our warranty.

### 7.3.1 Lower bellows replacement procedure

1 **Loosen the bellows mounting ring.**

2 **Pull out the bellows downward.**

3 **Mounting of the bellows is the reverse of removal.**

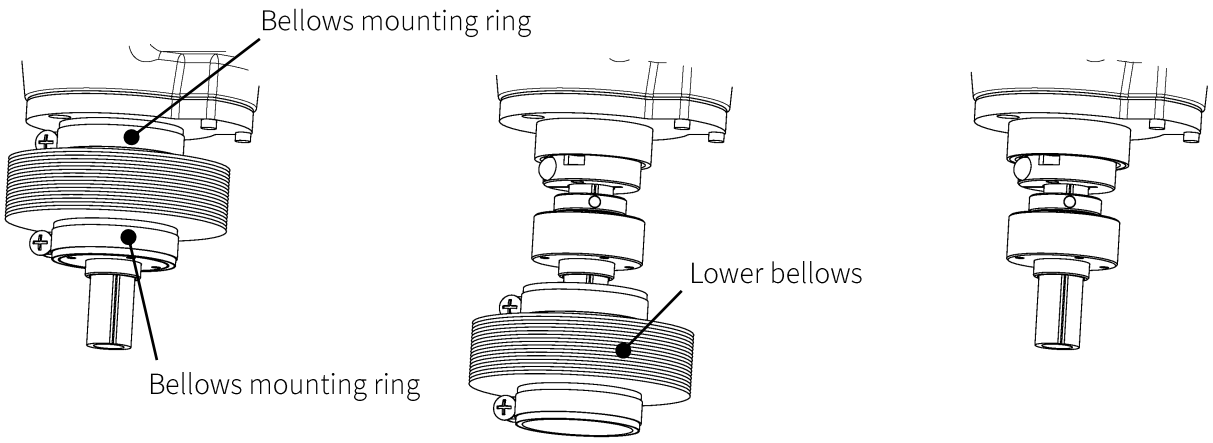



Fig. 7.2 Replacing the lower bellows

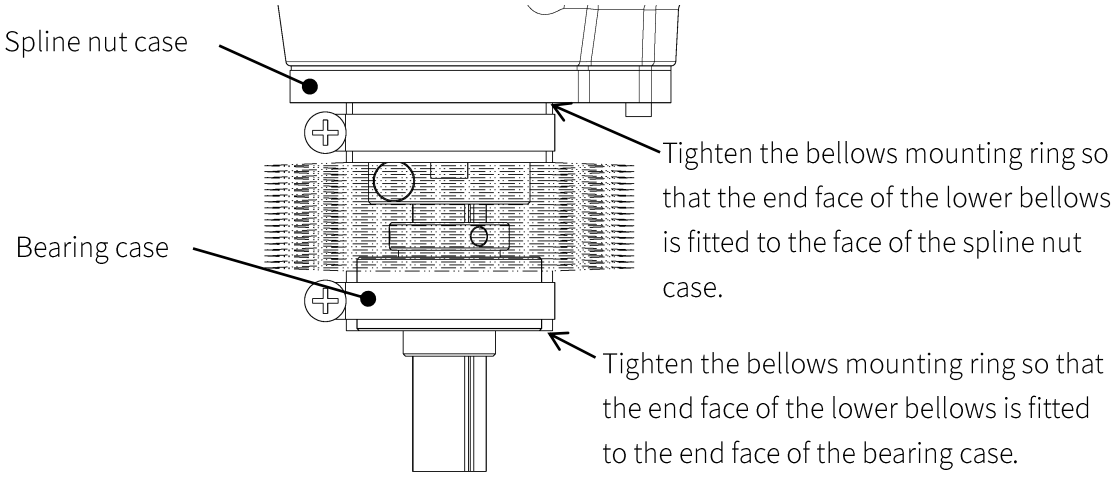



Fig. 7.3 Mounting position of the lower bellows

**NOTICE**

- When removing the bellows, do not pull the bellows forcedly.
  If the bellows is pulled forcedly, it will be torn.

### 7.3.2 Upper bellows replacement procedure

1 **Loosen the bellows mounting ring.**
2 **Remove the arm-2 cover. For removal of the arm-2 cover, see "7.4.2 Arm-2 cover."**
3 **Remove the lower presser plate and remove the bellows.**
4 **If the ball screw spline is also replaced, remove the two hexagon socket head cap screws securing the bellows shaft. For removal of the ball screw spline, refer to the Instruction Manual "STA1105 THE400 Robot Manual".**
5 **Mounting of bellows is the reverse of removal. Apply liquid gasket to the side face of the bearing case. Wipe off the protruding liquid gasket.**

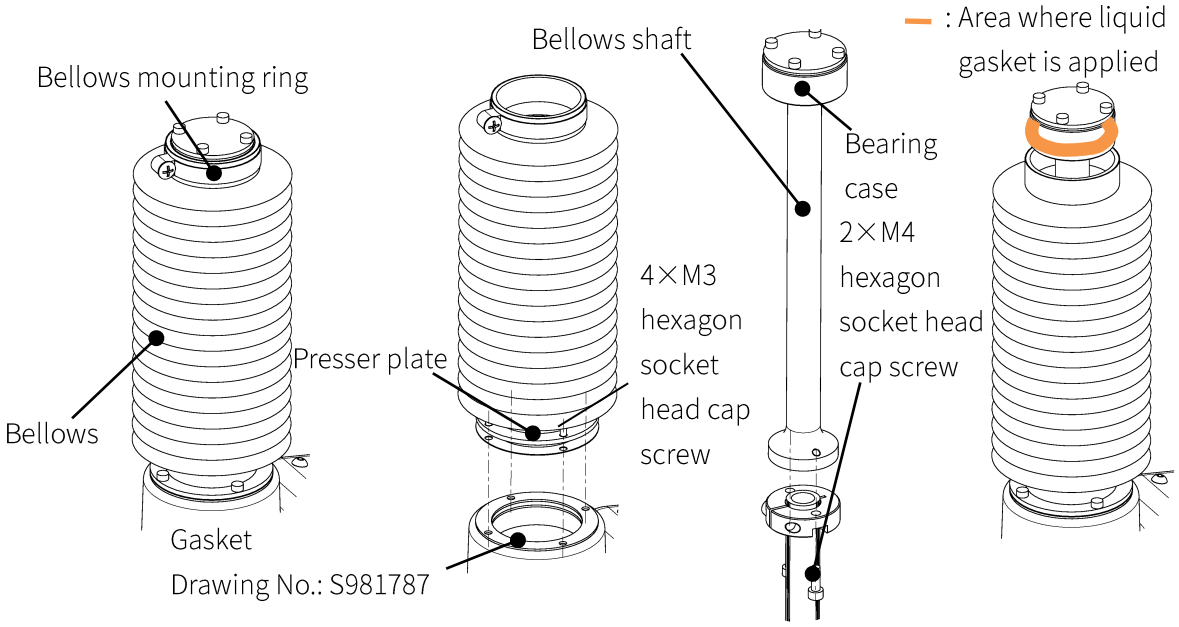



Fig. 7.4 Replacing the upper bellows

## NOTICE

- When removing the bellows, do not pull the bellows forcedly.
  If the bellows is pulled forcedly, it will be torn.
- Be careful not to damage the gasket.
  If the gasket is damaged, the dustproof and dripproof performance will be impaired to cause intrusion of dust etc.

*MEMO*

*The gasket tears easily and may be broken during parts replacement. It is therefore recommended to also prepare a gasket for replacement in the case of parts replacement. For information about the gasket, see "9.1 List of Maintenance Parts."*

### 7.3.3 Battery replacement procedure

Unlike the standard machine, the battery case of the dustproof and dripproof robot is covered with a cover. When replacing the battery, remove the cover as shown in Fig. 7.5. For the battery replacement procedure, refer to the Instruction Manual “STA1105 THE400 Robot Manual”.

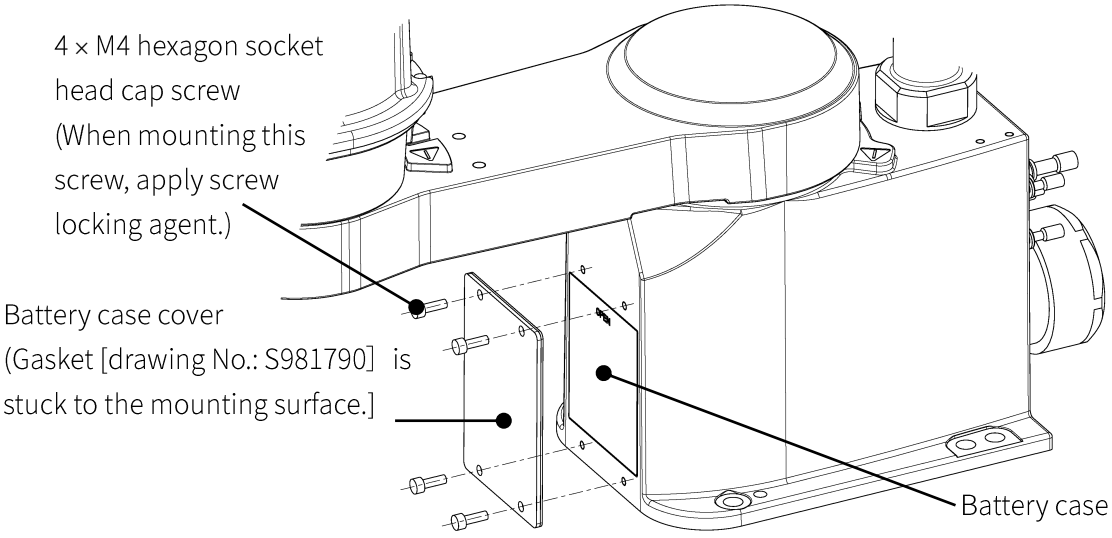



Fig. 7.5 Battery case cover

# NOTICE

- Be sure to apply screw locking adhesive to the mounting screws.
  Otherwise, the dustproof and dripproof performance will be impaired to cause intrusion of dust etc.
- Be careful not to damage the gasket.
  If the gasket is damaged, the dustproof and dripproof performance will be impaired to cause intrusion of dust etc.

*MEMO*

*The gasket tears easily and may be broken during parts replacement. It is therefore recommended to also prepare a gasket for replacement in the case of parts replacement. For information about the gasket, see “9.1 List of Maintenance Parts.”*

## 7.4 Removing and Mounting Covers

The dustproof and dripproof robot is different from the standard machine in that the gasket is mounted on the mounting surface of each cover. Strictly observe the work procedure in this section to do work.

**⚠DANGER**

- When opening the cover, be careful to prevent intrusion of moisture or foreign material into the robot. If the robot is energized with moisture or foreign material in it, electric shock will result.

**NOTICE**

- When opening the cover, be careful to prevent intrusion of moisture or foreign material into the robot. If the robot is energized with moisture or foreign material in it, the robot will become faulty.

### 7.4.1 Base rear cover

The base cover is secured to the base with hexagon socket head cap button screws (M4 × 8 × 6 pieces). When the fixing screws have been removed, the rear cover is separated from the base. However, since the cover is connected with the connectors inside, do not pull it forcibly. The gasket is stuck to the cover.
When mounting or removing the cover, be careful not to damage the gasket.

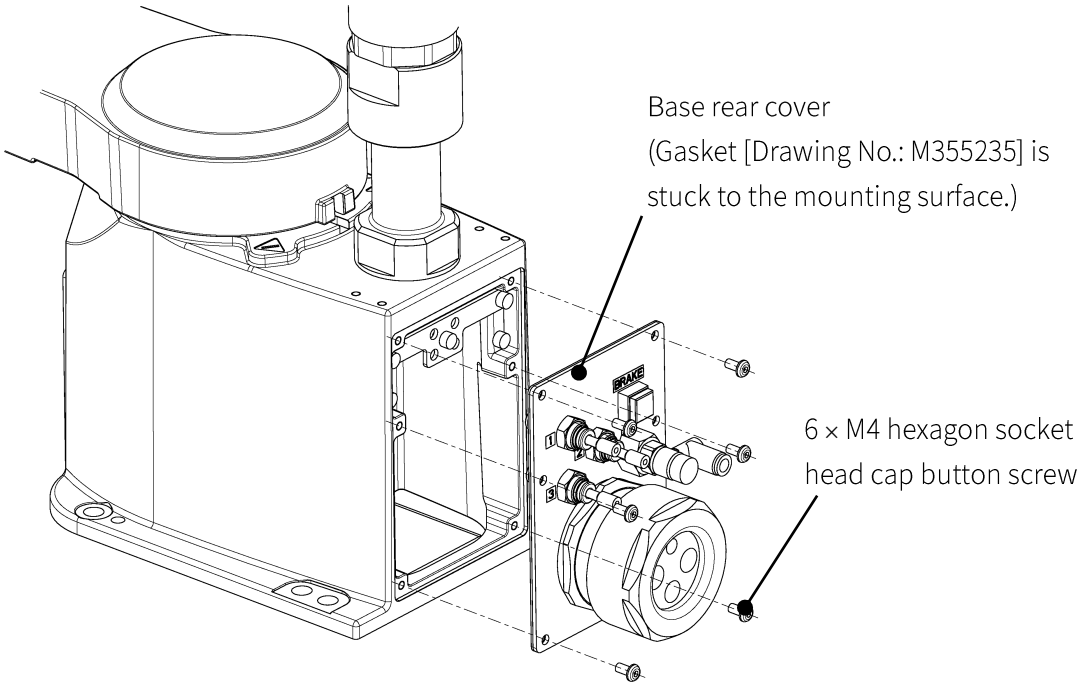



Fig. 7.6 Base rear cover

**NOTICE**

- Be careful not to damage the gasket.
  If the gasket is damaged, the dustproof and dripproof performance will be impaired to cause intrusion of dust etc.

*MEMO*

*The gasket tears easily and may be broken during parts replacement. It is therefore recommended to also prepare a gasket for replacement in the case of parts replacement. For information about the gasket, see "9.1List of Maintenance Parts."*

### 7.4.2 Arm-2 cover

The arm-2 cover is secured to arm 2 and the cover fixing guide with the hexagon socket head cap screws (M3 × 6 × 6 pieces), hexagon socket head cap screws (M3 × 8 × 2 pieces) and hexagon socket head cap button screws (M4 × 8 ×8 pieces).
The gasket is stuck to arm 2 and cover fixing guide. Be careful not to damage or peel the gasket forcedly. There is an area where the liquid gasket is partially stuck to the cover. If the gasket cannot be peeled well, use a flathead screw driver etc. to peel the gasket carefully so as not to damage the gasket.

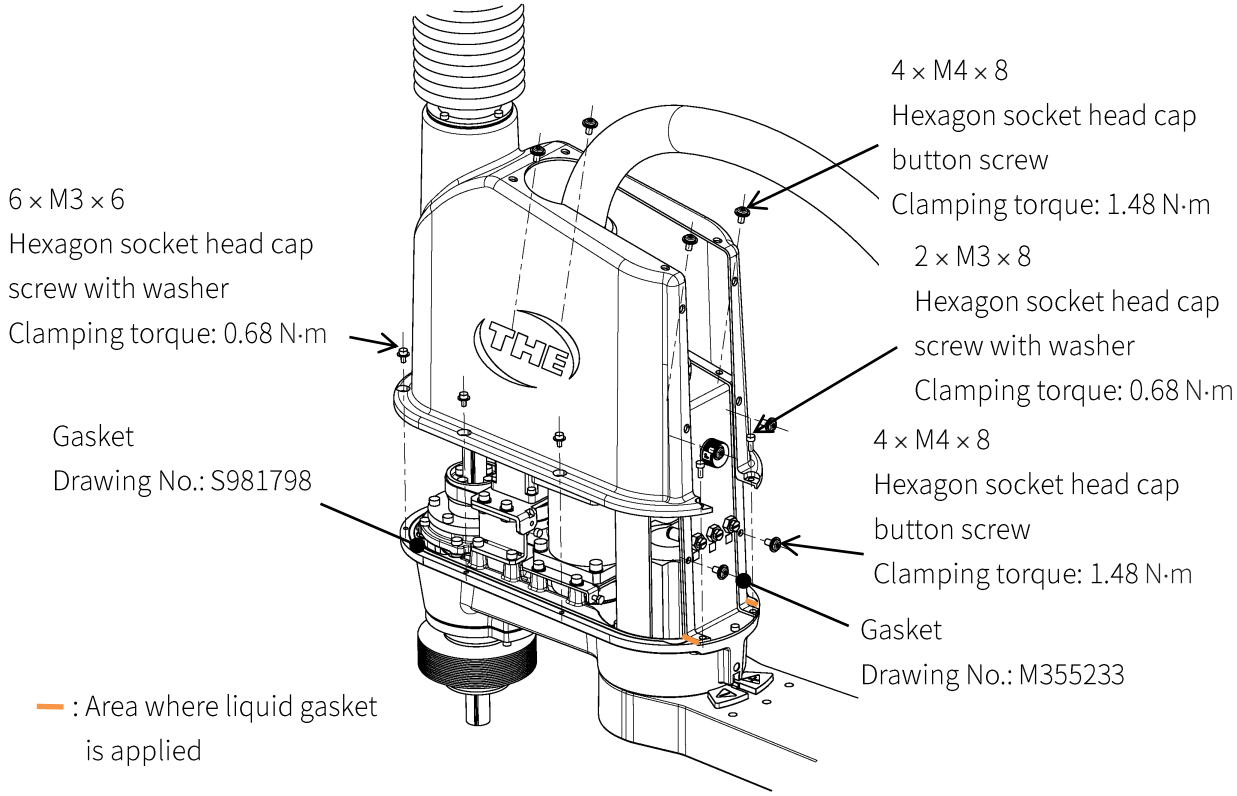



Fig. 7.7 Arm-2 cover

# NOTICE

- Be sure to apply screw locking adhesive to the mounting screws.
  Otherwise, the dustproof and dripproof performance will be impaired to cause intrusion of dust etc.
- Be careful not to damage the gasket.
  If the gasket is damaged, the dustproof and dripproof performance will be impaired to cause intrusion of dust etc.

*MEMO*

*The gasket tears easily and may be broken during parts replacement. It is therefore recommended to also prepare a gasket for replacement in the case of parts replacement. For information about the gasket, see "9.1List of Maintenance Parts."*

# 8. Robot Zero Point and Position Detector Error

This section describes the items that are different from those for the standard machine about the zero point of the dustproof and dripproof robot. For other items, refer to the Instruction Manual “STA1105 THE400 Robot Manual”.

## 8.1 Zero Point Posture When Multi-turn Data Is Cleared

The dustproof and dripproof robot is different from the standard machine in the height of axis 3 when multiturn data is cleared. See Fig. 8.1 to set the position of ball screw spline.

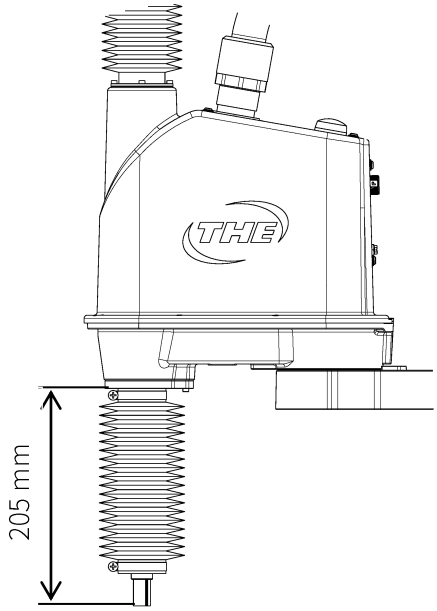



Fig. 8.1 Axis 3 setting position

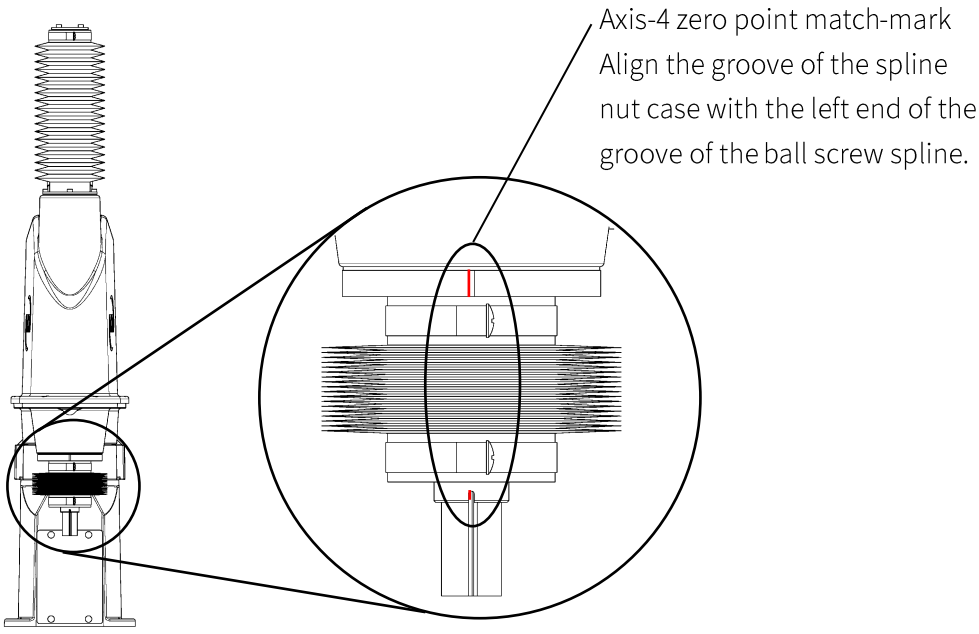



Fig. 8.2 Axis 3 setting position

# 9. Replacement Parts for Maintenance

This section describes different items about the replacement parts for maintenance of the dustproof and dripproof robot. For other replacement parts for maintenance, refer to the Instruction Manual “STA1105 THE400 Robot Manual”.

## 9.1 List of Maintenance Parts

| No | Part name | Model | Drawing No. | Unit code | Manufacturer | Q’ty | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Bellows | | S968253 | Y610D0FT0 | SHIBAURA MACHINE | 1 | Upper |
| 2 | | | S968254 | Y610D0FV0 | | 1 | Lower |
| 3 | Body harness | | L27871G01 | Y610D0G70 | SHIBAURA MACHINE | 1 | |
| 4 | Gasket | | S981787 | | SHIBAURA MACHINE | 2 | See 7.3.2 |
| 5 | Gasket | | S981790 | | SHIBAURA MACHINE | 1 | See 7.3.3. |
| 6 | Gasket | | S981798 | | SHIBAURA MACHINE | 1 | See 7.4.2. |
| 7 | Gasket | | M355233 | | SHIBAURA MACHINE | 1 | See 7.4.2. |
| 8 | Gasket | | M355235 | | SHIBAURA MACHINE | 1 | See 7.4.1. |

# 10. Disposal

## 10.1 Notes on Disposal

When disposing of the robot body, pallets and packing materials that have become unnecessary after the product is unpacked, parts that have become unnecessary after replacement, bundling tie cut during maintenance, waste cloth to which grease has adhered, etc., handle them as industrial waste.
When disposing of them, observe the regulations of your company. SHIBAURA MACHINE does not collect and dispose of them.

**⚠DANGER**

- Never burn, short-circuit, charge, disassemble, or heat the battery.
  If liquid that leads after the battery blows out adheres to your skin or eye, inflammation results.
  - When liquid adheres to your skin, wash it away with flowing water immediately.
  - When liquid adheres to your eye, wash it away with flowing water immediately and then see a doctor.
  - Dispose of the battery according to the regulations of your company.

**NOTICE**

- Be sure to dispose of the packing materials that have become unnecessary according to the regulations of your company.
  When storing the packing materials, be sure to manage them to prevent them from being used for other than packing the robot.